"Eco-Driven Innovations: Advancing Plastics, Circular Economy & Climate Action for a Sustainable Future"

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Message of the Vice Chancellor, The Open University of Sri Lanka

It is with great pleasure that I extend my warmest greetings to all participants of the International Conference on Plastics, Innovations, and Environmental Sustainability (ICPIES) 2025, organized by the Faculty of Engineering Technology of The Open University of Sri Lanka.

The theme of this year's conference *"Eco-Driven Innovations: Advancing Plastics, Circular Economy & Climate Action for a Sustainable Future"* underscores the urgency of our collective responsibility to safeguard the planet. It calls for developing innovative solutions, advancing scientific knowledge, and promoting responsible practices that can transform the way we produce, use, and recycle plastics.

While plastics are indispensable in modern life, they present immense challenges to ecosystems, human health, and climate resilience. Addressing these issues requires bold ideas, technological innovation and collaborative action. This timely conference unites researchers, academics, policymakers, industry leaders, and practitioners from across the country to deliberate on one of the most pressing challenges of our time.

As a university dedicated to advancing knowledge and fostering innovation, The Open University of Sri Lanka is proud to host a platform that not only encourages academic discourse but also inspires actionable solutions. By addressing plastic pollution, circular economy strategies, and sustainable technologies, this conference contributes to shaping the policies and practices essential for a greener and more resilient future.

I am confident that the outcomes of ICPIES 2025 will inspire meaningful collaborations and foster new ideas that extend well beyond this gathering. I take this opportunity to commend the Faculty of Engineering Technology for their vision and leadership in organizing this important event. I wish all participants an inspiring conference experience that broadens perspectives, strengthens collaborations, and brings us closer to a sustainable future.

Senior Professor P. M. C. Thilakerathne

Vice-Chancellor
The Open University of Sri Lanka
Sri Lanka

Message from the Dean of Faculty of Engineering Technology, The Open University of Sri Lanka

As innovators from diverse backgrounds unite to design a cleaner and brighter future, I am honoured to welcome all participants to the International Conference on Plastics, Innovations, and Environmental Sustainability (ICPIES 2025), hosted by the Faculty of Engineering Technology at The Open University of Sri Lanka.

This year's theme, "Eco-Driven Innovations: Advancing Plastics, Circular Economy & Climate Action for a Sustainable Future," responds to the global demand for responsible innovation. As engineers, scientists, and technologists, we have a responsibility to ensure that progress does not compromise environmental integrity. Eco-driven advances protect the planet while also promoting social equity and economic prosperity, generating benefits for all. At the Faculty of Engineering Technology, we are committed to preparing future engineers, technologists, and researchers with both technical expertise and a strong dedication to sustainability. ICPIES 2025 provides students and the academic community with opportunities to engage with international experts, pioneering research, and practical solutions that integrate science, technology, and society.

I extend my sincere gratitude to the steering committee, technical committees, distinguished speakers, and all contributors for their dedication to making this event possible. Their efforts have established an environment that fosters meaningful knowledge exchange and impactful collaboration.

I encourage all participants to utilise this platform to share research, build new networks, and establish collaborations that continue beyond the conference. To promote meaningful connections and co-creation, I invite each of you to identify and connect with at least one cross-disciplinary collaborator during ICPIES 2025. This initiative is intended to transform networking into actionable partnerships and foster lasting professional relationships. I wish everyone a valuable and rewarding experience at ICPIES 2025.

Dr. L.S.K. Udugama,

Dean
Faculty of Engineering Technology
The Open University of Sri Lanka
Sri Lanka

Messages from Conference Chairs

Prof. Jagath Manatunge

It is with great warmth and deep respect that I welcome you to the 2nd International Conference on Plastics, Innovations, and Environmental Sustainability (ICPIES 2025). This event brings together a community of scholars, professionals, policymakers, and students who share a common purpose: to advance knowledge and practice in addressing one of the most pressing challenges of our time.

The growing concern over plastics and their impact on our environment demands responses that are thoughtful, innovative, and collaborative. ICPIES 2025 provides a rare platform where ideas from diverse disciplines intersect, where evidence and experience inform debate, and where new partnerships can take root. It is in such spaces that solutions gain both scientific depth and practical relevance.

I wish to express my gratitude to the keynote and plenary speakers, the scientific and technical committees, the reviewers, sponsors, and above all, the authors and participants. Your dedication and contributions give this conference its true strength and meaning.

I encourage all of you to engage actively in the sessions, to exchange perspectives generously, and to seek opportunities for collaboration that endure beyond these two days. The success of ICPIES 2025 will be measured not only in the richness of the discussions but also in the collective resolve it fosters toward sustainable transformation.

With best wishes for a productive, collegial, and inspiring conference experience, I warmly welcome you to ICPIES 2025.

Prof. Jagath Manatunge

Dean
Faculty of Engineering
University of Moratuwa
Sri Lanka

Messages from Conference Chairs

Prof. Bandunee Athapattu

It is with great pride and enthusiasm that I extend a warm welcome to all participants of the 2nd International Conference on Plastics, Innovations, and Environmental Sustainability (ICPIES 2025). This conference, to be held on the 24th and 25th of September 2025 at the Cinnamon Lakeside Hotel, Colombo, is a significant event that brings together thought leaders, researchers, and innovators to address one of the most pressing global challenges: sustainability in the realm of plastics and environmental impact.

As the Conference Chair of the ICPIES 2025, I am particularly excited about this year's theme, "Eco-driven Innovations: Advancing Plastics, Circular Economy, and Climate Action for a Sustainable Future." The focus on eco-friendly innovations, circular economy principles, and climate action resonates deeply with our commitment to finding sustainable solutions that protect and preserve the environment. This year's conference underscores the urgent need for collaboration among institutions, industries, and governments to reduce plastic waste, promote circular economy practices, and foster sustainable development. The Open University of Sri Lanka is proud to be part of this important dialogue, particularly as we collaborate with the University of Moratuwa and other key stakeholders to drive meaningful research and innovation in these critical areas.

I would like to express my heartfelt gratitude to the organizing committee, speakers, participants, and all contributors for their invaluable efforts in making this conference a reality. The research and discussions that will take place over the next two days will provide critical insights into how we can continue to innovate and create solutions that foster environmental sustainability for generations to come.

I encourage everyone to actively engage, exchange ideas, and collaborate towards achieving our shared goals of a sustainable, eco-driven future. Together, we can make a tangible impact on the global movement toward sustainability and climate action. Thank you, and I wish you all a successful, productive, and inspiring conference.

Prof. Bandunee Athapattu

Professor in Environmental Engineering
Faculty of Engineering Technology
The Open University of Sri Lanka
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Keynote Address 1: Cement & Concrete Sustainability Roadmap / Towards Net-Zero Carbon Concrete by 2050

Concrete is the most widely used material on Earth after water. It builds our homes, our bridges, our hospitals, and the infrastructure that connects societies. But we also know it comes with a significant environmental footprint: cement and concrete together account for around 7–8% of global CO_2 emissions. This makes our sector not just a participant in climate conversation, but a key actor in delivering real solutions.

Reaching net zero by 2050 is not only an aspiration but also a necessity. To get there, we must pursue a roadmap defined by innovation, collaboration, and accountability. And it is possible, if we follow a clear roadmap:

- By 2030: Reduce emissions by at least 25–30% through clinker substitution, alternative fuels, and efficiency gains.
- By 2040: Deploy carbon capture use and sequestration (CCUS) at scale, achieve circular concrete systems, and shift to near-100% renewable electricity.
- By 2050: Deliver net zero concrete that supports global development while respecting planetary boundaries.

No company, no country, can achieve this alone. Success requires collaboration between producers, policymakers, academia, financiers, and society. Policy frameworks such as carbon pricing, cement &concrete carbon rating, green public procurement, and performance-based codes will accelerate the transition. Investors and customers must reward low-carbon solutions, creating markets that value sustainability.

The same material that has built civilizations for millennia can now contribute to preserve the planet for future generations. The roadmap to net zero concrete by 2050 is not just about cutting emissions. It is about reimagining our industry, reinventing our practices, and redefining our legacy.

If we act with urgency, with unity, and with innovation, we can ensure that when future generations walk through the cities and communities we have built, they will see not only concrete structures but also the enduring proof that our industry chose responsibility over complacency, and action over delay.

Dr. Moussa Baalbaki

Director Products & Solutions Portfolio
INSEE Cement

Keynote Address 2: Eco-Driven Innovations: Advancing Plastics, Circular Economy & Climate Action for a Sustainable Future

Distinguished guests, colleagues, practitioners, and researchers,

It is a great privilege to stand before you today at the International Conference on Plastics, Innovations, and Environmental Sustainability, hosted by the Open University of Sri Lanka. ICPIES 2025 comes at a time when the world is urgently seeking pathways to address plastic pollution and climate change together. What makes this conference particularly significant is that it bridges the perspectives of researchers and practitioners, ensuring that scientific insight translates into actionable solutions that work in the realities of our communities.

Sri Lanka provides both a sobering case study and a site of opportunity. Research shows that the country generates around 1.6 million metric tonnes of plastic waste annually, of which nearly 70 percent consists of single-use plastics. According to the National Plastic Waste Inventory, about 250,000 tonnes of plastic enter the waste stream each year, with an average of 11 kilograms per person, yet only around 11 percent of plastic in mixed waste is actually recycled. These statistics remind us that although our collection systems recover much of the waste, too little is effectively reused or reintegrated into a circular system. Leakage into the natural environment is particularly troubling. The recent study has identified Sri Lanka as the leading contributor in South Asia to plastic leakage into waterways, with the Kelani River alone releasing more than 6,000 metric tonnes of plastic into the sea each year. Coastal areas such as Galle, Mirissa, and Unawatuna, which are both biodiversity hotspots and vital to tourism, bear the visible scars of this crisis.

From the researcher's standpoint, these figures reflect structural inefficiencies in our material flows. A material flow analysis conducted in 2017 already warned that plastic consumption in Sri Lanka was growing by about 16 percent annually, with recycling efficiency at only three percent and collection efficiency around one-third. The same study projected that nearly 70,000 tonnes of plastic could enter the marine environment if trends continued. Today, those projections are not hypothetical, they are our lived reality. As academics, we therefore have the responsibility to not only generate knowledge but also to provide models and scenarios that can guide effective interventions.

From the practitioner's side, however, the challenges manifest differently. Regulations banning single-use plastics, such as thin polythene bags and plastic straws, are on the books, yet enforcement is uneven across regions. Retailers continue to supply banned items in places like Anuradhapura due to consumer demand, limited awareness, and the higher cost of alternatives. Recycling infrastructure, though improving, remains inadequate. Only a fraction of recyclable plastics find their way into formal recycling streams, with the majority handled by informal waste pickers or ending up burned, dumped, or leaked into waterways. Moreover, alternatives to plastics are often cost-prohibitive, limiting widespread adoption. Practitioners face the daily reality of fragmented institutional responsibilities, weak coordination among agencies, and gaps in financial resources.

This dual perspective highlights the gap between research and practice. Researchers identify the urgency and quantify the risks, while practitioners wrestle with enforcement, affordability, and the capacity to scale up innovations. Yet it also shows where the opportunities lie. Expanding material-flow modelling and lifecycle assessment can help predict the impact of policies such as extended producer responsibility (EPR) or source segregation at household level. Community-based systems, such as waste banks and school or fisher cooperative initiatives, demonstrate how source segregation can succeed if integrated into cultural and social practices. Affordable, locally developed alternatives, whether natural fiber packaging, reusable containers, or biodegradable materials, need the support of both research and policy to reach scale.

Technology also provides new frontiers. Monitoring tools, such as GIS mapping of waste leakage, AI-assisted litter detection, and remote sensors, can enable targeted interventions. Similarly, pilot projects in EPR, if designed with transparent accountability and strong data systems, can become templates for nationwide programs. At the same time, the social dimension must not be overlooked. Education and behavior change are critical. Research into what incentives shift consumer behavior, what messages resonate with different demographics, and how to engage communities must guide awareness campaigns. Practitioners then have the task of embedding those strategies into schools, local councils, and business practices.

Looking ahead, Sri Lanka has the potential to become a regional leader in circular plastics and climate action. Imagine a future where plastic is designed for reuse, recycling, or safe biodegradation from the start; where informal waste pickers are integrated into formal systems with dignity and recognition; where rivers and beaches are free of plastic pollution, supporting thriving ecosystems and tourism; and where innovation in materials and systems creates new green jobs. Realizing this vision requires measurable targets, stronger institutional capacity, sustained investment in infrastructure, and continuous collaboration between researchers, policymakers, industry, and communities.

In conclusion, the evidence is clear: plastic waste in Sri Lanka poses serious risks to the environment, health, and economy. The lived experience of practitioners shows us that while progress has been made, the gap between policy and practice remains wide. The role of ICPIES 2025 is to close that gap, to connect data to action, innovation to implementation, and vision to reality. I urge all of you to share your knowledge, challenge assumptions, and forge partnerships that can outlast this conference. Together, let us ensure that Sri Lanka not only responds to the plastic crisis but leads in advancing a truly circular and climate-resilient economy.

Thank you, and I wish you an inspiring and fruitful conference.

Vijayapala Sinnathamby

Sri Lanka Programme Manager
United Nations Office for Project Services (UNOPS)

Address by the Guest of Honour: A Decade of Bilateral Scientific Cooperation for Safe Water and Public Health

Established in 2015 under the witness of the Presidents of China and Sri Lanka, the China–Sri Lanka Joint Research and Demonstration Center for Water Technology (JRDC) was conceived as a strategic response to Sri Lanka's urgent needs for safe drinking water and public health solutions, especially in rural and Chronic Kidney Disease of Unknown Etiology (CKDu)-affected regions. As a national-level international cooperation platform, JRDC exemplifies the commitment of both nations to strengthening scientific collaboration and delivering tangible benefits to local communities.

Research and Demonstration complex was completed in 2020 with the total investment exceeded 4 billion Sri Lankan Rupees, funded through China Aid and became fully operational in 2021 at the University of Peradeniya. JRDC is jointly managed by the Sri Lankan Ministry of Urban Development, Housing and Construction, the Chinese Academy of Sciences (CAS), and the University of Peradeniya. Its operation reflects the spirit of "joint consultation, joint construction, and joint benefit." JRDC's mission aligns with the Sustainable Development Goals (SDGs), particularly SDG 6: Clean Water and Sanitation, and contributes to broader regional goals of environmental sustainability, public health security, and inclusive innovation. The center is equipped with advanced laboratories, pilot plants, and training facilities, as well as specialized instruments for water quality analysis and environmental monitoring, making it the most advanced water science facility in Sri Lanka and South Asia.

JRDC has made significant contributions to safe water supply, scientific research, human capacity development across Sri Lanka. More than seven demonstration systems using advanced technologies such as Electrodialysis Reversal (EDR) and Nanofiltration (NF) have been installed in CKDu-affected regions, providing clean drinking water to over 6,000 villagers. These systems have successfully reduced fluoride and nitrate levels, ensuring that treated water meets Sri Lankan national standards (SLS 614:2013) and directly strengthening community health and resilience. At the same time, JRDC has advanced research on groundwater contamination, environmental risk factors for CKDu, and safe water technologies, with over 100 peer-reviewed international publications co-authored by Chinese and Sri Lankan scientists.

JRDC has also hosted more than 10 international workshops & bilateral academic exchanges. More than 40 Sri Lankan scholars have obtained PhD or Master's degrees through CAS/ANSO fellowships and over 70 professionals in water management and public health have received technical training through JRDC programs in Sri Lanka. Center representatives have participated in global platforms such as COP27 and the IWA WDCE, and High-Level Conferences. The center also supports student internships, laboratory programs, and university collaborations. JRDC's achievements have gained national and international recognition, including the "Best Water Engineering Project" award from the Ministry of Water Supply in 2020. Further, GRDC maintains active collaboration with institutions such as the Board of Investments (BOI), the National Water Supply & Drainage Board (NWSDB), the Central Environmental Authority (CEA), the Department of National Community Water Supply (DNCWS), and regional universities positioning itself as a key hub for international water science diplomacy.

Ongoing project under JRDC includes,

- Sweet Water Rural Water Supply Project, deployment of the solar-powered water treatment units, providing a replicable model for decentralized rural water solutions
- Kelani River Protection, for pollution source tracing and community engagement activities to improve surface water quality in the Kelani River basin
- Eco-Industrial Park Standards which supports the localization of international Eco-Industrial Park standards in Sri Lanka through pilot projects and policy alignment
- Science communication and awareness campaigns on water safety in CKDu-affected districts, in collaboration with local schools and public health workers
- The Youth Sustainability Environmental Leadership (USEL) Program, launched in 2025, empowering over 500 Sri Lankan undergraduates to take leadership roles in sustainable development through training, competition, and international collaboration

Aligned with Sri Lanka's 2024 National Science and Technology Policy, JRDC focuses on six national R&D priorities, especially environment, land, water, air, and mineral resources. Through joint research, technology localization, and capacity building, JRDC aims to enhance institutional and professional capacity in the following areas,

- Rural Drinking Water Safety
- Climate-Resilient Integrated Water Resource Management
- Eco-Industrial Park Development
- Air Monitoring and Pollution Control

The sustainable development of JRDC will depend on strengthening governance structures, improving operational efficiency, and establishing diversified funding mechanisms to ensure its long-term viability. Through high-level and sustainable collaboration, JRDC aims to contribute to long-term resilience, shared prosperity, and mutual benefit between the China & Sri Lanka.

Prof. Wang Yawei

Director
Office of the Overseas Centre for Environmental Science & Technology Cooperation
Chinese Academy of Sciences

Plastic Pollution and Health Impacts: Microplastics in Terrestrial & Marine Ecosystem

Investigation of Microplastics in Commercially Available Salt in Sri Lanka

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Microplastic (MP) contamination in sea salt is a growing concern linked to marine plastic pollution. The necessity to evaluate MPs in commercially available salt in Sri Lanka is highlighted by the absence of comprehensive data on MP contamination and potential intake by salt consumption. Powdered and crystalline salt from ten local brands, were randomly collected with one imported sample from the USA. A solution of 100 g of salt was prepared in ultrapure water and treated hydrogen peroxide and incubated for organic matter digestion. The samples were filtered using the Whatman GF/C membrane, and initially observed under a digital microscope. Nile Red staining was employed to quantify MP under UV light, and Fourier Transform Infrared spectroscopy was performed to identify the polymer type. Data from a community-based survey of 44 individuals were used in an exposure analysis. All salt samples were contaminated with MPs with an average abundance of 523.125±143.049 particles/kg in the crystal form and higher levels of MP abundance of 836.154±139.853 particles/kg in the powdered form. The predominant morphologies were fibers and fragments, with fibers indicating a size range of 0.212–4.617 mm for all particles. Polyethylene terephthalate and polyvinyl chloride have been confirmed in black and blue domain colors due to their usage in packaging materials. The estimated average daily intake (EDI) of MPs from salt consumption was 2.022 ± 0.553 particles/day from crystalline salt and 3.233 ± 0.541 particles/day from powdered salt. This is proportional to the estimated average annual intake (EAI) of MPs, 738.176 ± 201.855 particles/year with crystalline salt and 1179.888 ± 197.346 particles/year from powdered salt. These results emphasize the requirement of quality control in sea salt production. The baseline data for health risk assessments can be obtained by this initial EDI and EAI in Sri Lanka, with further research recommended to assess long-term impacts.

Keywords: *Estimated Daily Intake, Food Safety, Health Risk Assessment, Human Exposure, Microplastic*

Effect of Regional Water on Microplastic Release from Tea Bags in Sri Lanka

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The rapid expansion of construction activities worldwide has led to a significant increase in construction waste. It poses substantial challenges to environmental sustainability and economic stability. For effective construction waste management, accurate quantification of onsite waste generation and a clear understanding of the factors contributing to waste generation are required. This study investigates material waste generation across different contractor grades in Sri Lanka, with the following objectives: (1) to quantify and compare onsite material waste generation for key building materials among various contractor grades; (2) to identify and analyse the principal factors influencing onsite construction waste, with a focus on contractor grade differences; and (3) to review prevailing waste management practices among Sri Lankan contractors and propose methodological improvements for future waste reduction. A questionnaire survey was conducted to collect data from multiple contractor grades. Both subjective and statistical analyses of variation were used to examine the relationships. The findings revealed differences in waste percentages across contractor grades. Specially in the subjective analysis, particularly for key materials such as concrete, sand, aggregate, steel, plastic, and wood. However, statistically, there is no significant difference in total construction waste generation among the various contractor grades. It suggests that all grades contribute comparably to overall waste output. The study identified material handling inefficiencies, design-related issues, and inadequate site management as major contributors to waste. The upper-grade contractors demonstrated more advanced waste management practices. Still, specific improvements need to be made to minimise waste generation in the industry.

Keywords- *Estimated daily intake, Microplastics, Plastic packaging, Water hardness*

ALDFG and Single-Use Plastics Based Marine Pollution Prevention in Sri Lanka: A Case Study from Rockland Corporate Responsibility

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Abandoned, Lost, or otherwise Discarded Fishing Gear (ALDFG) and single-use plastics are two major categories of persistent marine debris adversely affecting Sri Lanka's coastal ecosystems. This study presents findings from a countrywide, multi-site marine pollution prevention initiative conducted between 2022 and 2025 across 18 coastal zones of Sri Lanka. This action, coordinated by Rockland Corporate Responsibility (RDLCR) environmental program in partnership with youth and conservation organizations, included systematic shoreline and underwater debris removal, ecological monitoring, community mobilization and school-based environmental education. A total of 4.9 metric tons of non-biodegradable marine debris were collected, with 2,027 kg of ghost fishing gear retrieved from near shore line habitats and coral reef ecosystems. Spatial analysis indicated that significant amounts of debris concentrations were found in areas with river outflows and unregulated public beaches, particularly during the southwest monsoon. Biodiversity assessments recorded direct damage to coral colonies and entanglement of marine fauna, while mangrove habitats exhibited plastic accumulation and sedimentation anomalies. The program also highlights the participation of over 300 women in daily coastal waste collection and engaged youth in awareness-building across multiple districts. The findings signify the need for a national policy framework incorporating Extended Producer Responsibility (EPR), Extended Consumer Responsibility (ECR) and ghost gear management under marine conservation mandates. The case provides a scalable, evidence-based model for addressing marine plastic pollution through coordinated, multi-stakeholder interventions in island nation contexts.

Keywords: *ALDFG, Corporate Responsibility, Marine Pollution Prevention, Single use plastic,*

Microplastic Emission Dynamics in an Urban River under Rainfall Events

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Urban rivers are key pathways for land-sourced microplastics (MPs) to the oceans. However, their real-time emission dynamics during rainfall events remain poorly understood. This study presents high-frequency monitoring of MPs in an urban river in Japan over three rainfall events (light, moderate, and heavy). The event mean concentrations (EMCs) of MPs were influenced by the number of antecedent dry days, while total loads linked with the amount of rainfall. During moderate and heavy rainfall events, microplastic (MP) loads increased by up to 100 times and event mean concentrations (EMCs) by up to 350 times, compared to dry weather. MPs smaller than 2000 µm displayed first-flush effect during light and moderate rainfall, with over 50% of their total mass released within the initial 20–40% of runoff flow. The smallest particles (10–40 µm) were mobilized rapidly even at low rainfall intensities, whereas larger MPs (>2000 µm) were discharged around or after the peak rainfall. The polymer composition of riverine MPs was significantly impacted by rain, with polymer characteristics suggesting a likely origin from local road dust. The levels of MPs and total suspended solids (TSS) in river water during rainfall were well-correlated, suggesting the possibility to use TSS measurements as a useful indicator for predicting MP levels. Finally, the findings highlighted the importance of targeting moderate to heavy rainfall events and intercepting stormwater first flush for effective control of MP pollution in urban rivers.

Keywords: *First flush, Microplastics, Temporal variations, Urban runoff,*

Microfiber Removal in Wastewater Treatment: A Stage-wise Analysis from Sri Lanka

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Microfibers, a major category of microplastics, have been widely detected in aquatic environments worldwide, posing threats to the natural environment. Wastewater treatment plants (WWTPs) are considered as a major sink and a source for microfibers. They are mainly found in the industrial wastewater derived especially from textiles. However, in Sri Lanka, the understanding of the removal of microfibers in WWTPs is limited, and thereby, the development of effective microfiber remediation strategies has been constrained. To address this knowledge gap, this study aims to assess the microfiber abundance at major treatment stages of a WWTP and the stage-wise microfiber removal efficiency, highlighting the temporal variations over four consecutive sampling months. Sampling was conducted in a centralized industrial WWTP, in the Colombo District, Sri Lanka, from January to April 2025, at six sampling locations, including inlet, equalization tank, inlet and outlet of the oxidation ditch, settlement tank (secondary), and discharge point, covering major treatment stages of the WWTP. Microfibers were visually quantified using stereomicroscopy. Results indicated a progressive microfiber reduction along the successive treatment stages with influent microfiber concentrations ranging from 29.11 ± 5.34 to 46.44 ± 11.47 microfibers/L. The secondary sedimentation tank demonstrated the highest removal achieving 78.5 %, serving as the major settling process of the WWTP, in the absence of primary settling tank. Microfiber concentration differed significantly across treatment stages (Kruskal-Wallis, $p = 0.003$) and was strongly negatively correlated with treatment stages (Spearman's $\rho = -0.843$). The final discharge point showed < 5 microfibers/L, highlighting an effective but incomplete removal. This study underscores the need for advanced treatment strategies for mitigating microfiber contamination in Sri Lankan wastewater treatment systems.

Keywords: *Industrial wastewater, microplastics, removal efficiency, wastewater treatment plant*

Microfiber Shedding in Domestic Laundering Using Washing Machines: What Does Multivariate Analysis Reveal?

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Microplastic pollution is increasingly recognized as a significant environmental threat, with synthetic microfibers from domestic laundry contributing notably to aquatic and terrestrial contamination. During each wash cycle, synthetic textiles shed microplastics, which enter wastewater systems via washing machines and subsequently accumulate in ecosystems, posing risks to both aquatic organisms and human health. In Sri Lanka, research quantifying microplastic emissions from household laundering remains limited. This study addresses this gap by quantifying and characterizing microplastics released from domestic washing machines in selected households within the Western Province, Sri Lanka. A total of 25 households were sampled, with 1 L of laundry effluent collected from each site. Samples were filtered and analyzed to determine microplastic levels based on shape, color, size, and polymer type. FTIR Spectroscopy was used to identify polymer composition. A structured questionnaire survey captured household washing practices and potential factors influencing microplastic release at the sampling point. Relationships between washing parameters and microfiber shedding were further examined using multivariate statistical techniques, including Principal Component Analysis (PCA) and Cluster Analysis (CA). The results showed a mean of 1456.7 ± 1294.8 microplastic particles per liter of laundry effluent, with fibers constituting 95% of the total count. Other observed forms included fragments (44.1/L), films (34/L), pellets (0.7/L), and flakes (1.2/L). The mean microplastic weight per sample was 0.1 ± 0.1 g, with a mean size ranging from 0.6 ± 0.2 mm. Color analysis revealed a predominance of black (32%) and blue (31%) microplastics. PCA and CA indicated that fabric type, washing machine specifications, detergent use, and washing conditions were the key influencers of microfiber shedding. These findings conclude that domestic laundering is a major source of microplastic pollution in Sri Lanka, highlighting the urgent need for improved washing practices, effective filtration technologies, and targeted policy measures to reduce environmental contamination.

Keywords: *Domestic laundering, Fourier Transform Infrared Spectroscopy (FTIR), Microplastic pollution, Multivariate statistical analysis, Synthetic microfibers*

Occurrence of Microplastics in Selected Liquid Oral Medications and Intravenous Infusions in Sri Lanka

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Microplastics (MPs) are increasingly recognized as emerging contaminants in clinical environments, suggesting novel exposure routes beyond dietary and inhalation routes. This study provides the first systematic investigation of MP contamination of selected intravenous (IV) fluids and oral liquid medications in Sri Lanka. Eleven medicinal liquids (four IV fluids and seven oral preparations) were collected from government and private healthcare facilities based on their frequency of usage and analyzed under strict quality control conditions in a laminar hood facility. Microplastics were extracted by vacuum filtration through 0.45 µm cellulose nitrate filters and identified by stereomicroscopy and hot needle testing. All samples contained MPs, with normal saline IV fluids showing a mean concentration of 15.30 ± 3.28 MPs per 100 mL and oral liquids 69.98 ± 43.35 MPs per 100 mL. Fragments and fibres were the dominant morphologies, consistent with global reports linking MPs to packaging and handling. These findings highlight a critical gap in pharmaceutical quality control, particularly considering the direct systemic exposure risk posed by IV infusions. Establishing monitoring systems, enhancing packaging standards, and formulating regulatory frameworks are urgently needed to mitigate MP associated in healthcare products as emerging contaminants, posing direct threats on human.

Keywords: *Contamination, Oral liquid medications, Microplastics, Medicinal liquids, Intravenous fluids,*

Cytogenotoxic effects of Polyethylene terephthalate (PET) microplastics on *Allium cepa* root tip cells: Implications for abnormal cell proliferation

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Microplastic (MP) pollution is an emerging environmental concern with ecological and human health implications. This study investigated the cytogenotoxic effects of polyethylene terephthalate (PET) MPs (<20 µm) on *Allium cepa* root tip cells. Root tips were exposed to varying concentrations of PET MPs (50, 100, 250, 500, and 1000 ppm) alongside untreated controls (n=10 for each condition, with 3000 cells analyzed per sample). Mitotic indices and cell death percentages were quantified using standardized microscopic techniques. Results demonstrated a dose-dependent relationship between PET MP concentration and cellular responses. Mitotic indices were 0.013±0.007 (control), 0.024±0.001 (50 ppm), 0.299±0.001 (100 ppm), 0.136±0.001 (250 ppm), 0.254±0.056 (500 ppm), and 0.543±0.006 (1000 ppm), indicating substantial alterations in cell division rates compared to control conditions. Moreover, cell death percentages increased proportionally with MP concentration: 1.0±0.2% (control), 6.4±0.2% (50 ppm), 20±0.4% (100 ppm), 23±0.6% (250 ppm), 25±0.6% (500 ppm), and 37±0.5% (1000 ppm). Statistical analysis using Kruskal-Wallis one-way ANOVA revealed significant differences among treatment groups ($p<0.05$) regarding cell death percentages. Further, macroscopic observations indicated longer root lengths in PET exposed bulbs compared to normal control. The observed pattern of increasingly abnormal cell proliferation with higher PET concentrations suggests these MPs may disrupt normal cell cycle regulation. This study provides novel evidence that PET MPs may induce abnormal mitotic activity and cellular damage in plant test systems, potentially inducing mitosis processes via mechanisms that warrant further investigations. These findings suggest scientific validations regarding the capacity of PET MPs to induce hyperproliferation patterns similar to those observed in early carcinogenesis, highlighting the need for comprehensive risk assessment and regulatory measures to address MPs contamination in environmental and agricultural systems.

Keywords: *Allium cepa, Microplastics, Polyethylene terephthalate, Proliferation*

Policy and Regulatory Frameworks for Waste Management

Plastic Waste Policies and Informal Waste Pickers: A Policy Gap Analysis in Sri Lanka

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The rapid increase in plastic waste has made effective plastic waste management a pressing necessity in Sri Lanka. Informal waste pickers contribute significantly to this process by collecting, sorting, and recycling a substantial portion of plastic waste. Despite their critical role, the extent to which national policies recognize and support these workers remains unclear. This study investigates the inclusion or exclusion of informal waste pickers within Sri Lanka's plastic waste management policies, with a focus on the presence of livelihood protection measures, institutional mechanisms for participation, and potential risks of penalization. Using content analysis of existing policy documents, the findings reveal a lack of formal recognition or support for informal waste pickers. The study highlights the urgent need for policy reforms, including amendments to the National Action Plan on Plastic Waste Management and the integration of financial compensation mechanisms under Extended Producer Responsibility (EPR) frameworks. By addressing these gaps, Sri Lanka can advance towards a more inclusive, efficient, and sustainable plastic waste management system that aligns with global best practices and social equity goals.

Keywords: *Informal Waste Pickers, Plastic Waste Management Policy, Sustainable Development*

Onsite Construction Waste Generation in Sri Lanka: Patterns, Causes, and Policy Implications

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The rapid expansion of construction activities worldwide has led to a significant increase in construction waste. It poses substantial challenges to environmental sustainability and economic stability. For effective construction waste management, accurate quantification of onsite waste generation and a clear understanding of the factors contributing to waste generation are required. This study investigates material waste generation across different contractor grades in Sri Lanka, with the following objectives: (1) to quantify and compare onsite material waste generation for key building materials among various contractor grades; (2) to identify and analyse the principal factors influencing onsite construction waste, with a focus on contractor grade differences; and (3) to review prevailing waste management practices among Sri Lankan contractors and propose methodological improvements for future waste reduction. A questionnaire survey was conducted to collect data from multiple contractor grades. Both subjective and statistical analyses of variation were used to examine the relationships. The findings revealed differences in waste percentages across contractor grades. Specially in the subjective analysis, particularly for key materials such as concrete, sand, aggregate, steel, plastic, and wood. However, statistically, there is no significant difference in total construction waste generation among the various contractor grades. It suggests that all grades contribute comparably to overall waste output. The study identified material handling inefficiencies, design-related issues, and inadequate site management as major contributors to waste. The upper-grade contractors demonstrated more advanced waste management practices. Still, specific improvements need to be made to minimise waste generation in the industry.

Keywords: *Construction waste, Contractor grade, Material waste estimation, Waste management*

Assessment of Municipal Solid Waste in Kaduwela, Sri Lanka: Composition Analysis and Policy Implications for Waste Management

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Municipal solid waste has become a huge challenge in Sri Lanka specially in urban areas. Kaduwela Municipal Council area is facing increasing pressure due to rapid urbanization, population growth, and rising living standards. Ineffective waste management practices have resulted in notable environmental degradation. Therefore, it emphasizes the growing necessity of sustainable and efficient approaches. This research focuses on evaluating the volume and types of municipal solid waste generated in Kaduwela, and attention is given to investigate relationship between household income levels and the quantity of waste produced. A self-completion questionnaire and field observations were used over a one-month period to find out the waste management practices. SPSS and Excel were used for data analysis. According to the findings three main disposal methods such as municipal collection (50%), home garden pits (26%), and a mix of both (24%) were found. While 96% of respondents reported awareness of waste separation, 80% engaged in composting, and 64% burned some of their waste. There is a clear relationship between monthly income and waste generation. Higher income households generated more paper, plastic, polythene, and rubber waste, while lower income households produced more food, glass, metal, and other organic waste. These results indicate the importance of income-based waste management planning. Even though public awareness is relatively high, well organized support system has not been implemented yet. To address this, new technical and financial strategies should be implemented by local authorities. Programs such as “pay-as-you-throw” could be implemented to encourage responsible waste disposal. Incentives for home compost preparation, Waste separation before dumping etc. along with strict enforcement of illegal dumping regulations, can promote better practices. Therefore, it needs better public engagement and institution support to improve solid waste management in order to reduce environmental impacts specially in rapidly urbanizing areas.

Keywords: *Income-level analysis, Household, Urban waste Waste composition, Waste disposal practices*

Bridging the Legal Gaps in Plastic Waste Governance in Sri Lanka's Regulatory Framework

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Plastic waste has become a major environmental and public health issue in Sri Lanka, made worse by rising Urbanization, Consumerism, and insufficient regulatory oversight. However, the country has enacted several Legal instruments, including the National Environmental Act No. 47 of 1980 and several gazette notifications banning plastic items. The existing Legal regime remains fractured, poorly enforced, and institutionally disconnected. This paper examines the shortcomings of Sri Lanka's Legal policy framework on plastic waste management and evaluates the gaps for its effectiveness. Using comparative analysis on India's plastic waste management rules and the European Union's Single-Use Plastics Directive. This paper suggests actionable reforms that could enhance legal clarity, enforcement Mechanisms, and sustainable practices in Sri Lanka's context using a qualitative and doctrinal approach supported by both primary and secondary sources, with legal and policy recommendations intended to align national plastic waste governance with international environmental standards and Sustainable Development Goals.

Keywords- *Plastic waste, waste regulation, legal reform, Environmental Law*

Repurposing Plastic Waste into Circular Production: A Model for Sustainable Textile Manufacturing

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The global textile industry is a major contributor to environmental degradation, particularly through plastic waste generated from packaging, chemical containers, and production processes. Teejay Lanka PLC, Sri Lanka's first multinational textile manufacturer, has addressed this challenge through an innovative plastic waste management initiative aligned with its Abhivarah 2030 sustainability roadmap. By adopting the 10R circular economy principles, Teejay has implemented a comprehensive strategy to reduce, reuse, and recycle plastic waste, with a special focus on converting hazardous chemical barrels into usable infrastructure components. A key breakthrough was the introduction of high-performance plastic tubes from recycled chemical barrels to serve as the central core for fabric rolls. These recycled tubes underwent rigorous testing for mechanical strength and chemical safety, meeting the standards set by the AFIRM Group's 2025 Packaging Restricted Substances List. Produced tubes demonstrated performance comparable to virgin materials and passed all regulatory checks for heavy metals, phthalates, APEOs, and PAHs. This initiative has led to a substantial reduction in virgin plastic use and improved operational efficiency. Furthermore, Teejay's success has strengthened its brand reputation, fostered community partnerships, and established a scalable model for circular innovation within the textile sector.

Keywords: *Circular Economy, Plastic Waste Management, Sustainable Manufacturing, Textile Industry*

Alternatives for Plastics -Challenges and Opportunities

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Alternatives to plastics are needed due to the severe environmental and health problems associated with their production, use, and disposal. Plastic pollution, particularly in oceans, is a major concern, and traditional plastics don't easily biodegrade, leading to long-term environmental damage. Many plastics are not easily recyclable, leading to increased landfill waste and potential environmental contamination. Additionally, plastic production relies heavily on fossil fuels, contributing to greenhouse gas emissions and climate change. Lack of technical skills for managing hazardous waste, insufficient infrastructure development for recycling and recovery, and above all, lack of awareness of the rules and regulations are the key factors behind this massive pile of plastic waste. The severity of plastic pollution exerts an adverse effect on the environment and total ecosystem. In this study, a comprehensive analysis by introducing and replacing alternatives for plastics, as well as its effect on the human being and ecological system, is discussed in terms of source identification with respect to developed and developing countries. Moreover, this study sheds light on sustainable waste management procedures and identifies the key challenges to adopting effective measures to minimize the negative impact of plastic alternatives for plastics. The breakdown of plastics into micro plastics can contaminate food and water sources, potentially impacting human health. Economically Waste Management Costs the cost of managing plastic waste, including collection, sorting, and disposal, can be substantial, especially for developing countries lacking adequate infrastructure. Biodegradable Plastics made from renewable resources like cornstarch or sugarcane can be broken down by microorganisms under specific conditions, reducing reliance on fossil fuels and potentially minimizing pollution.

Keywords: *Algae based materials, Bio-plastics, Cellulose based materials, Hemp-based products, Mushroom-based materials, Recycled materials*

Technological Innovations in Sustainable Environment

Design and Fabrication of Sustainable Flat Washers Using Textile Fabric Waste

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Sri Lanka's vibrant textile and apparel industry generates a substantial amount of textile waste, particularly discarded denim, from its manufacturing processes. Both large-scale production facilities and small garment workshops contribute to this accumulation, making denim waste an abundant, yet largely untapped, resource and a significant pollutant. Currently, most discarded denim ends up in landfills, where synthetic dyes and chemical finishes contaminate ecosystems. To mitigate this environmental impact, this research investigates the feasibility of designing and fabricating sustainable flat washers using a denim fabric waste composite. The composite material was prepared by embedding denim fabric waste into an epoxy resin matrix, which was then cross-linked using a methyl ethyl ketone peroxide catalyst. Different denim fiber-to-resin ratios (1:15, 1:20, 1:25, and 1:30) were prepared according to ASTM standards to investigate their mechanical properties, including compressive strength, tensile strength, flexural strength, and hardness. The 1:20 denim fiber-to-resin ratio composite consistently exhibited superior mechanical properties compared to the other composites. The 1:20 denim to resin ratio composite showed the flexural strength of 356.67 MPa which suggests the material's capability to withstand bending loads effectively. The tensile strength of 5.33 MPa was achieved by 1:20 fiber to resin composite indicating sufficient durability for tension-based applications and also exhibited compressive strength of 29.47 GPa highlights the material's ability to resist high compressive forces, making it suitable for heavy-load applications. Although their hardness is lower compared to conventional washers made from aluminum and mild steel, they still provide adequate hardness for many applications. The findings underscore the potential of using recycled textile waste as a viable alternative to traditional materials, offering environmental benefits without significantly compromising mechanical performance. This makes the 1:20 ratio composite flat washers a promising option for applications where sustainability is a key consideration, such as induction motor holder, fan head screw, automobile engine holding place and bicycle rear wheel axle washers. This work contributes to Sustainable Development Goals (SDGs) 9 and 12 in Sri Lanka.

Keywords: *Composite materials, Denim waste, flat washers, Mechanical properties, Sustainable manufacturing*

Unrefined Montmorillonite Nanoclay-Cinnamon Wood Biochar Composite for Simultaneous Removal of Hardness and Fluoride in Groundwater

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It was aimed to synthesize an adsorbent for fluoride and hardness removal using a nanoclay-biochar composite from unrefined Montmorillonite (MMT) clay from the Murunkan in Mannar, Sri Lanka. Fourier Transform Infrared Spectroscopy (FTIR), X-Ray Diffraction (XRD), X-Ray Fluorescence (XRF), Scanning Electron Microscope (SEM), and Thermal Gravimetric Analysis (TGA) were employed to characterize the materials. FTIR, XRD, and TGA analyses indicated the presence of MMT clay mineral in the sample. The SEM image of the composite confirmed the successful binding of MMT onto the biochar through a flaky appearance along the porous morphology. Further, the surface texture of clay and biochar could support hardness and fluoride removal by enhancing adsorption. The results obtained from CHN analysis show that the biochar sample was rich in carbon, which helps in the sorption of hardness and fluoride. The adsorption isotherm data of all the adsorbents were well fitted to the Temkin isotherm model, suggesting that the removal of hardness and fluoride onto the adsorbent indicates a heat of sorption. The kinetic adsorption data of all the adsorbents were well fitted to the pseudo-first-order kinetic model, indicating the dominant contribution of the physisorption mechanism. However, since the measured adsorption was minimal, a modification to the nanoclay-biochar composite is required for higher sorption efficiency by enhancing the active sites on sorbents to purify groundwater that contains an undesirable level of selected contaminants.

Keywords: *Adsorption, Biochar, Montmorillonite nanoclay, Murukkan clay, Nanocomposite*

Design a Retro-Reflective Eco-Friendly Clay Rooftile to Reduce the Urban Heat Island (UHI) Effect

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The Urban Heat Island (UHI) effect, where the urban areas experience higher and warmer temperatures compared to surrounding rural areas is currently becoming a serious climatic phenomenon common to a lot of large cities around the world. This rise in urban temperatures lead to an increase in energy consumption of buildings and moreover, thermal discomfort to people and affects the quality of human life. There have been reported cases that this high temperature has adversely affected infants, elderly people and people with poor health conditions. Several strategies have been proposed and developed to overcome the UHI effect. This study is focused on developing a rooftile as a mitigation strategy for UHI. Sieve analysis was done and 1.18mm to 0.6mm waste glass particle size was selected as the optimum particle size for designing the reflective layer. Then rooftile was designed by applying the reflective layer, RR glass bead layer and protective sheet and Retro-reflectivity testing was done to check the RR properties of developed Rooftiles. The temperature around Wayamba Royal College, Kurunegala, Sri Lanka has been analyzed in the current study. It is located in a highly crowded area with high relative humidity and temperature. Field temperature data was collected from 15th of April to 19th of April using a pre-prepared electronic setup. The developed rooftile has reduced the surface temperature by 11.81 °C in the peak temperature hour of 11a.m to 12 noon on 19th of April. Therefore, this study explains the concept of UHI and the efficiency of RR rooftiles in mitigating the UHI effect.

Keyword: *Cool Rooftiles, Retro Reflective, Urban Heat Island*

Solidified Water Treatment Sludge as Potential Supplementary Cementious Material in Pervious Concrete Pavers for Mitigating Heat Island Effect

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Rapid urbanization is increasing impervious surfaces and urban heat. This study explores lightweight pervious concrete pavers with water treatment sludge and/or ceramic dust. Paver mixes were made with different ratios of sludge, where 10% replacement of fine aggregate with oven-dried sludge was still structurally sound and practical. While calcined sludge pavers had much higher compressive strength than the other pavers, the oven-dried sludge gave much less energy and cost inputs. Thermal testing of the pervious pavement compared to conventional concrete showed a 6°C drop in bottom-surface temperature at a top temperature of 40°C, though the effect diminished as temperature increased. These results confirm that sludge-based pervious concrete pavers effectively manage heat and retain structural and hydraulic performance for real-world applications and urban infrastructure challenges.

Keyword: *Ceramic Powder, Heat Island, Permeability, Upcycling*

Closed-Loop Copper Recovery: Integrating Electrolytic with Circular Economy Practices

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This study presents an economically viable and environmentally friendly technology to recover copper from the waste copper sulfate ($CuSO_4$) solution that is generated in electrolytic refining processes. One of the most significant concerns is the industrial effluent high copper content exceeding acceptable discharge limits and causing major environmental hazards. The primary objectives were to determine such things as the type of waste $CuSO_4$ solution that was, how much Fe was required for iron (Fe) and a maximum value of copper recovery, compared with such other metals like aluminum how effective it worked, and purity measurement.

We did batch experiments with different amounts of Fe, analyzed the results, filtered them, dried them, and used SEM and EDAX to describe them. For stepwise Fe addition, the optimal Fe dosage was determined to be 3 g per 50 mL of solution that resulted in solid copper with a purify of 43.54 wt % and the copper content in the solution decreased from: 63.33 g/L to 8.085 g/L and higher recovery efficiency as well as shorter process times were achieved when using single-step as compared to stepwise Fe addition.

The process offers significant economic advantages due to the low cost and wide availability of iron, minimal chemical inputs, and simplicity of operation. The process helps in resource recovery, pollution reduction and essentially follows the principles of circular economy by converting hazardous effluents to reusable metal. The alloy aluminum was tested as a potential standing in reducing agent, but no reaction could be observed here either, which seems to prove that iron is the better species for this application. In summary, iron cementation is a versatile and sustainable process according to these laboratory and industrial runs.

Keywords: *Circular Economy, Copper Recover, Iron Cementation, Sustainable Metallurgy, Wastewater Treatment*

The Mechanical Properties of PET Fiber Reinforced Concrete

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The widespread use of plastic products in everyday life has increased significantly in recent years. The resulting large volume of plastic waste has raised environmental concerns, particularly regarding its disposal. Recycling offers a practical solution to manage plastic waste without overburdening landfill sites. One such approach is the incorporation of plastic waste into concrete production. This review evaluates the mechanical performance of concrete reinforced with polyethylene terephthalate (PET) fibers derived from plastic waste. It summarizes studies that utilize PET bottles as fibers in concrete and provides details on the effects of PET on workability, compressive strength, splitting tensile strength, and flexural strength. Key findings indicate that appropriate PET fiber content can significantly enhance mechanical properties: compressive strength improved by up to 35.0% at 0.8% PET (compared to 5.7% for 0.1% polypropylene), splitting tensile strength increased by 81.0% at 0.4% PET (versus 15.8% at 0.3% polypropylene), and flexural strength increased by over 120.0% at 0.4% PET (compared to 17.2% at 0.3% polypropylene). However, the workability of concrete decreased by 24.0% at 0.8% PET and dropped by 78% at 1.0% PET. In conclusion, incorporating recycled PET fibers into concrete is not only a promising method for improving mechanical properties but also an environmentally friendly solution that reduces plastic waste and the construction industry's environmental footprint.

Keywords: *Compressive strength, Fiber concrete, Flexural strength, Polyethylene terephthalate, PET fiber, Splitting tensile strength*

Development of a Sustainable and Skin-Friendly Marigold Flower Inspired Resort Wear Collection Using Natural Dyeing Techniques

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Clothing, often referred to as our second skin, holds immense significance in human existence, undergoing continuous evolution throughout history. Herbal textiles are shown special properties in clothing that is used for a variety of purposes. This research investigates the potential of design and develops a resort wear collection for Sri Lankan ladies in age group 22-26, inspired with Marigold flower for spring and summer 2023/24. Natural dye of the Marigold flower has vibrant hue and medicinal properties was used as a value addition for fabric surface preparation and it benefits for cure skin allergies. Natural herbal fabric dyes are gaining popularity due to their eco-friendly and skin-friendly properties. More than the use of Marigold flower for concepts and design developments, this study examines the properties of Marigold flowers and their potential medicinal benefits for skin health. Due to its abundance of antibacterial agents, anti-inflammatory chemicals, and antioxidants, Marigold is a good choice for people with sensitive skin types and allergies. Designed and developed resort wear collection with free size silhouettes, highlighted with unique features of Marigold flower and offer comfort and skin care for the wearer encouraging taking part in leisure activities at resorts. Combining the benefits of natural herbal fabric dye with free-size resort wear collections presents a holistic approach to sustainable and skin-friendly fashion. This design approach not only captures the essence of nature-inspired elegance but also paves the way for potential technological innovations that promote sustainability in contemporary fashion and clothing practices.

Keywords: *Marigold flower, Natural dye, Resort wear collection, Sustainable clothing*

Green Synthesized Nanoparticle-Based Sustainable Filter Material for Enhanced Wastewater Treatment

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In the developing nations, most of the wastewater-generating industries fall into the medium scale, and these industries are predominantly treated with primary wastewater treatment methods with linear economic significances, with a lack of pollutant removal capabilities and limited sustainability. This paper introduces a sustainable filter material with zero-waste and circular economy concepts. Filter material composed of green synthesis of iron oxide nanoparticles (IONPs), activated carbon (AC), and montmorillonite clay. IONP (α-Fe_2O_3) synthesis was conducted by co-precipitation using Aloe Vera leaf extract as a green mediator, and composite filter material was fabricated by manual mixing of IONPs, AC, and clay. The synthesized filter material demonstrates a 58.5% reduction in chemical oxygen demand (COD) over 24 hours, a 43.2% COD reduction over 7 hours, a 27.8% color removal over 7 hours, and a 27.3% dissolved phosphate removal within 7 hours, achieved under practical laboratory conditions by optimizing flow rate and contact time. Additionally, enhancing contact time improves the filter material removal efficiency. The proposed zero-waste strategical wastewater treatment system with novel filter material improves the biological degradation spontaneously in the wastewater system and increases the capacity of removal of organic matter and nutrients from the wastewater. The discharged filter material reused strategies of plant growing media and seed storage capsules for used filter materials to support the wastewater treatment system for adherence to circular economy significances by diverging from the linear economy concepts.

Keywords: *Alpha-Hematite, Circular Economy, Nanotechnology, Sustainable, Zero-waste*

An Investigation of the Influence of Microplastics on the Thermal Conductivity of Seawater

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Plastic pollution has been the most concerning topic all over the world in the last decade. This plastic debris breaks down into smaller fragments, which are known as microplastics and nanoplastics. Even though the terrestrial environment, aquatic environment and air are contaminated with microplastics and nanoplastics, the marine environment is highly contaminated by microplastics and nanoplastics. The identification and elimination of microplastics and nanoplastics from the marine environment is very important. Since several conventional methods used to detect microplastics and nanoplastics are high-cost and limited to nanoplastics. Thermal analysis ensures the detection of microplastics and nanoplastics due to the unique thermal properties of polymers. This study focuses on determining the thermal behavior of polyethylene (PE) microplastics in artificial seawater. The thermal conductivity of artificial seawater and PE microplastics spiked with artificial seawater at the concentrations of 40 and 80 mg/L were measured using a Flucon LAMBDA thermal conductivity meter within a temperature range of 28 – 70℃. Scanning Electron Microscope (SEM) analysis of pristine PE microplastics and PE microplastics exposed to thermal conductivity analysis were performed using Zeiss EVO LS15 SEM. PE microplastics were decreased the thermal conductivity of artificial seawater in the temperature range 28 – 40℃. Due to the thermal insulation property of PE microplastics, it decreased the thermal conductivity. Meanwhile, PE microplastics increased the thermal conductivity of artificial seawater in the temperature range 40 – 70℃. PE microplastics were fragmented to smaller particles above 40℃ and the fragmented particles were increased the thermal conductivity due to Brownian motion. SEM images were confirmed that fragmented particles formed. This study proved that thermal conductivity analysis would be a potential detection method of microplastics and nanoplastics in the aquatic environment.

Keywords: *Microplastics, Nanoplastics, Thermal property, Seawater*

Upcycled Sustainable Evening Wear for Sri Lankan Women: Inspired by the Elegance of the Turkey Tail Mushroom

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Fashion is a captivating aspect of society, allowing individuals to express their uniqueness and enhance their attractiveness. This project transforms discarded clothes into extraordinary designs, inspired by the resilience and ecological importance of turkey tail mushrooms. These mushrooms, thriving on decaying wood, symbolize longevity and the connection between life and the environment. Guided by the theme "Be great with confidence," the project aims to heal the environment through upcycling. Designs incorporate the mushrooms' curved lines and vibrant colours, translated into delicate ruffles echoing their graceful forms. Techniques like fabric manipulation, texture development, colouring, and draping turn discarded textiles into sustainable, fashionable pieces. Targeting urban women aged 25 to 35, the Spring/Summer 2023/24 evening wear collection uses only upcycled fabrics, blending eco-consciousness with style. This approach offers unique fashion while fostering sustainability in the apparel industry and paving the way for technological innovations, promoting eco-friendly practices.

Keywords: *Eco-friendly, Fabric Manipulation, Sustainability, Upcycling*

Sustainable Braided Beach Resort Wear for Sri Lanka: Natural Fabrics and Eco-Friendly Dyeing Innovations

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This study explores the transformation of food waste into a sustainable resource for textile design while reviving traditional craft practices. Discarded coffee grounds, red cabbage leaves, marigold petals, avocado pits, and tea leaves were repurposed as natural dyes for biodegradable fabrics such as cotton, silk, and rayon. The research focused on developing the Braided Resort Wear Collection, inspired by the Lime Butterfly (Papilio demoleus), aligning creative vision with environmental responsibility. An online survey of over 100 participants assessed awareness and interest in sustainable fashion, natural dye use, and garments featuring traditional craftsmanship. Results showed strong support for eco-conscious clothing that combines environmental innovation with cultural authenticity. Dyeing trials evaluated colour uptake, fastness, and pH sensitivity, showing excellent results on natural fibres but reduced performance on synthetic blends. Collaboration with local women artisans skilled in eco-dyeing, braiding, and Beeralu lace-making enhanced garment quality, preserved Sri Lanka's textile heritage, and supported artisans' economic empowerment. To showcase the design potential, a six-piece Bohemian-inspired resort wear capsule, Flutter Weave, was produced. The project embodies SDG 12—Responsible Consumption and Production—by demonstrating how waste valorization, responsible sourcing, and artisan-led production can unite environmental stewardship with cultural richness. This approach highlights the aesthetic and functional potential of natural dyes while laying a foundation for future innovations in sustainable fashion and textiles.

Keywords: *Braided resort wear, Bohemian style, Sustainable clothing, Natural dye*

A Low-Cost IoT Platform with Packet Relaying for Sustainable Environmental Monitoring

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Rapid advancement of technology has created a new set of needs and increased the complexity of human lives. Technology offers efficient solutions to address these evolving challenges while prioritizing human safety and well-being. Among these emerging solutions, the Internet of Things (IoT) stands out for its ability to interconnect devices to collect and exchange data seamlessly. Remote sensing for outdoor environments, a critical application of IoT, has significant potential to enhance our understanding and management of natural and urban ecosystems. However, currently, there is a lack of readily available platforms that users can easily customize to suit their specific sensors or measurements. This project addresses this gap by developing a versatile and easily customizable IoT platform for outdoor remote sensing. The proposed system operates independently of third-party infrastructure, like GSM (Global System for Mobile Communication), ensuring functionality in rural areas. By leveraging LoRa technology, the platform's nodes can cover a 640 m-diameter cell, with packet relaying further extending coverage without the need for additional gateways. The system's nodes can maintain local backups in case of disruption and achieve a battery life of 60 days with a 19.24 Wh battery through the usage of low-power modes in the ATmega328p microcontroller. Flexibility is a key feature of the platform, with nodes supporting interfaces for two analog inputs, two SPI (Serial Peripheral Interface) devices, and two I2C (Inter-Integrated Circuit) devices, allowing for diverse sensor integration. Collected data is visualized through a customizable Power BI dashboard, enabling users to tailor the output to their specific analytical needs. This comprehensive platform offers a robust solution for outdoor remote sensing, combining remote communication, energy efficiency, and versatile data collection capabilities to meet the evolving challenges of environmental monitoring and management.

Keywords: *Environmental Monitoring, Internet of Things (IoT), LoRa Technology, Remote Sensing*

A Simplified Vision-Based Vehicle Speed Estimation Algorithm on FPGA

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While vision-based speed detection is a powerful tool for intelligent transportation systems, its adoption is often limited by the high computational and bandwidth costs of centralized processing. This research addresses that gap by proposing a highly efficient, hardware-accelerated solution implemented on a Field Programmable Gate Array (FPGA). The core of our design is a simplified vision-based algorithm that calculates speed by analyzing pixel data from only two virtual intrusion lines. This approach innovatively bypasses computationally expensive 2D image processing, requiring no object detection or segmentation, which is key to its minimal latency and power consumption. To validate our design, we prototyped the architecture on a Zynq-7000 SoC and evaluated its performance against the BrnoCompSpeed benchmark dataset. Our software simulations achieved a competitive mean absolute error of 2.4 km/h. The hardware implementation proved to be exceptionally resource-efficient, utilizing less than 1% of the available logic and operating at 100 MHz. This demonstrates the feasibility of our approach and offers a clear path toward mass-producible, intelligent sensors that can make traffic enforcement more affordable, responsive, and widespread. Crucially, the low power consumption inherent to an FPGA solution offers a sustainable alternative to power-intensive GPU-based systems for large-scale ITS deployments.

Keywords: *FPGA implementation, Real-time vision processing, Vision-based vehicle speed detection, Intelligent transportation systems*

Convertible Ladies' Casual Wear Collection (S/S) for the U.S. Inspired by the Pearl Oyster

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This study presents the design and development of a convertible, comfortable casual wear collection for upper-class single women aged 25–30 in the United States. It addresses the lifestyle demands and psychological needs of this demographic, shaped by a fast-paced, career-driven culture and often conservative social attitudes. Many in this group prioritize professional success, face societal undervaluation, and have limited time for leisure or personal well-being. There is a clear demand for versatile garments that transition effortlessly between professional and social settings. Comfort here is both physical and psychological, supporting confidence and ease. The collection integrates functional, convertible design elements to meet these needs. Its conceptual framework draws from the film Miss Congeniality (2000), whose protagonist reflects independence, resilience, and adaptability qualities central to the target consumer. Design inspiration comes from the pearl oyster, symbolizing transformation, understated beauty, and strength. Developed under the Tommy Hilfiger brand, known for youthful and innovative aesthetics, the collection uses lightweight linen for comfort, breathability, and seasonal suitability. Elegant embroidery and convertible features allow easy adaptation for varied occasions, merging functionality with refined style.

Keywords: *Casual wear, Convertible, Dynamic lifestyle, Spring & Summer, US*

Clay Biochar Composites in Treating Laboratory Wastewater

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Laboratory wastewater contains hazardous physical and chemical pollutants which results detrimental impact to the human, animals and environment. Horizontal flow constructed wetland with clay - biochar composite, Calicut tile, biochar, clay, wetland plants, 20mm aggregates by natural resources available at nearby places. Laboratory Wastewater collected from the collection tank diluted to desirable pH with pipe borne water was allowed to pass through the treatment system. Slightly higher pH increase observed during passage through the bed due to the filter media of biochar's alkalinity nature. Higher removal efficiencies of 80% BOD and 89% COD observed in the treatment system may due to the presence of biofilm in final treatment tank and clay: biochar composite as filter media. Gasified cinnamon biochar by-product of syngas production, plays a key role in reducing COD by 86% in the proto type model due to its distinctive physiochemical properties such as higher BET SSA, morphology of SEM analysis, lower H/C & higher O/C molar ratio in CHN analysis. Higher removal efficiency of BOD and COD observed for horizontal flow constructed wetlands constructed by utilizing onsite available materials and clay - biochar composites. The dispersed small laboratories can benefit more from running an economical and ecologically friendly treatment system with locally accessible natural materials.

Keywords: *Biochar, Clay biochar composite, Constructed Wetland, Heavy metal*

Performance Evaluation of Glass Fiber Reinforced Pervious Concrete for Urban Infrastructure

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Pervious concrete (PC) is increasingly recognized as a sustainable pavement material due to its ability to reduce stormwater runoff and improve groundwater recharge. However, its widespread application is limited by relatively low strength and poor durability. This study investigates the combined use of alkali-resistant glass fibers (ARGF) and silica fume (SF) to improve the mechanical performance of PC while maintaining its permeability. ARGF, with high zirconia content for alkali resistance, was incorporated to enhance crack resistance and tensile properties, while SF, a highly reactive pozzolanic material, was added to refine pore structure and strengthen the cementitious matrix. Experimental testing included compressive strength, split tensile strength, porosity, and permeability assessments at curing ages of 14 and 28 days. Results indicated that incorporating 1.5% ARGF by cement weight yielded a 52% increase in split tensile strength compared to the control at 28 days. The addition of SF further improved matrix density and strength without compromising PC's open-cell structure. Observations of failure planes revealed that cracks propagated through the cement paste rather than aggregates. These findings demonstrate that the combination of ARGF and SF enhances both the durability and tensile performance of PC, enabling its broader application beyond light-load pavements. The study provides a practical approach for producing high-performance pervious concrete with improved mechanical properties, contributing to sustainable pavement design and long-term urban infrastructure resilience.

Keywords : *Alkali Resistance, Glass Fiber, Pervious Concrete*

A Comprehensive analysis of grid power quality issues in the proposed electrification of Panadura - Veyangoda

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Railway electrification presents significant opportunities for addressing urban traffic congestion while introducing complex power quality challenges that require careful analysis and mitigation. The proposed electrification of Sri Lanka's Panadura-Veyangoda railway corridor, serving 44% of national railway passengers across 64 kilometers, necessitates a comprehensive power quality assessment to ensure grid stability and operational reliability. The research utilized a two-tier simulation framework using ETAP 2018 for steady-state analysis and MATLAB Simulink 2023a for dynamic traction modeling, examining critical time scenarios with 10 Electrical Multiple Units representing maximum electrical demand conditions. Two substations, Ragama and Dehiwala, were modeled, outfitted with 12.5 MVA transformers to service 5.2 MW Electrical Multiple Units at a 0.9 power factor. Results revealed voltage drops of 0.33% at 132 kV grid interconnection points during steady-state conditions, within IEEE Std 519-2014 limits. But dynamic analysis showed voltage reductions reaching 43.93% during simultaneous train acceleration, exceeding the acceptable 10% thresholds. Harmonic analysis indicated total harmonic distortion levels of 7.92% at grid interconnection points, primarily from 3rd (3.74%), 5th (1.98%), and 7th (0.54%) order components generated by variable frequency drive systems. Passive harmonic filters significantly improved power quality by reducing Total Harmonic Distortion by 74% through targeted attenuation of 3rd, 5th, 7th, and 11th harmonic orders. The study offers crucial technical groundwork for Sri Lanka's railway electrification, including validated simulations and effective strategies for maintaining reliable electric railway performance.

Keywords: *Electric railway, Harmonic distortion, Power quality, Railway electrification, Traction substation, Voltage regulation*

Microgrid for Low Voltage Distribution Systems and a Model for Optimum Power Sharing

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Grid-connected microgrids have become financially and technically feasible due to the high integration of renewable-based distributed generators and the fast development of battery technology. The formation of microgrids facilitates the integration of multiple sources of energy including renewables such as solar and wind at the distribution level. Since the solar irradiation level in Sri Lanka is sufficient for electricity generation using solar photovoltaics throughout the year, the paper proposes forming microgrids for existing low voltage feeders utilizing solar PV as one of the main micro-sources of energy for the microgrids. Other than solar PV, the proposed microgrid consists of a grid supply, battery energy storage device, and diesel generator. Within this study, an algorithm has been developed to optimally share the energy demand of the low voltage distribution system between utility grid, solar photovoltaics, battery energy storage device, and diesel generator. Based on the proposed algorithm, microgrid simulation software has been developed in the MATLAB-Simulink environment. Formula for the determination of most suitable capacities of micro-sources and battery energy storage device were proposed. The proposed model was tested to transform an existing low voltage distribution system into a microgrid. Optimum power dispatch between the sources of the microgrid during 24 hours of a day has been presented using the developed software. The results show that depending on the time of the day the use of solar PV, battery energy storage device, and diesel generator have more financial benefits than the use of grid energy alone throughout the day.

Keywords: *Distribution systems, microgrid, Rooftop Solar PV*

Review on; The Heavy Metal Filtration Efficiency of Mangrove Root Systems

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Mangrove forests are unique coastal ecosystems that play an important role in reducing pollution, especially from heavy metals. These pollutants, including lead (Pb), cadmium (Cd), copper (Cu), and zinc (Zn), often enter coastal waters from industries, aquaculture, and urban development. Mangrove roots act as major sinks for Pb, Cd, Cu, and Zn, with metals concentrated in roots rather than aerial parts, reducing ecological risks. Avicennia marina effectively immobilizes Zn and Cu, Rhizosphere microdata accumulates higher Cu and Zn, while Kandelia candel shows strong Cd and Pb retention. Sediment properties such as grain size, organic matter, and redox potential regulate metal availability, and ecological studies confirm mangroves also act as bio indicators of contamination severity. Heavy metal retention in mangroves happens through several processes such as trapping sediments, changes in soil chemistry around roots, and natural barriers in roots that block metals from moving upward. Laboratory studies explain how metals are stored at the cellular level, while field studies show that factors like salinity, water flow, and seasonal changes affect how metals behave in real ecosystems. Studies from around the world, including Sri Lanka, show that the ability of mangroves to filter metals depends on soil chemistry, tidal flows, and human activities. Protecting and restoring mangrove ecosystems is therefore vital for sustainable coastal management, bioremediation, and long-term environmental safety.

Keywords: *Bioaccumulation, Coastal Pollution, Filtration efficiency, Heavy metals, Mangroves*

Comparative Evaluation of Bioplastic Films from Invasive Water Hyacinth and Banana Peel Waste: An Alternative to Plastics for Circular Economy Integration

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Sustainable, environmentally friendly alternatives to petroleum-based plastics are increasingly needed. This study investigates the potential of converting two common organic waste materials in Sri Lanka, water hyacinth plants [Salvinia molesta (SM) and Pistia stratiotes (PS)] and banana peels [Musa spp.] into biodegradable bioplastics. Cellulose was extracted from sun-dried and oven-dried stems of SM and PS. Starch-rich pulp was prepared from overripe Musa spp. These were blended with gelatin, glycerol, and acetic acid to create biodegradable films using a heat mixing method. Physical, chemical and environmental properties of synthesized bioplastics were analysed. The combination of Salvinia molesta and Musa genus showed the best performance. This film had a smooth surface, strong chemical bonding, excellent heat resistance (over 120 °C), low moisture content (0.23%), and high biodegradability (99% in two days). However, the Pistia stratiotes and Musa spp. formulation had poor structure, absorbed more water, had low density, and degraded more slowly. A survey of 100 people in the Colombo District showed that good public awareness (64.4%) of bioplastics and a willingness to pay more (57.7%) for them, although cost and limited availability were significant concerns. This study explored the conversion of local waste into valuable, eco-friendly materials, addressing both environmental and social needs

Keywords: *Biodegradability, Eco-friendly materials, Public perception*

Advancing Sustainability through Material Innovation and Circular Designs in Plastics

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The transition toward a circular economy in plastics hinges on rethinking both the materials we use, and the way plastic products are designed, produced, and managed throughout their life cycle. Conventional plastics, while versatile and cost-effective, pose major sustainability challenges due to their fossil-based origins, persistence in the environment, and low recycling efficiency. To address these concerns, researchers and industries are increasingly turning toward material innovations such as bio-based, biodegradable, and chemically recyclable plastics, which offer the potential to reduce dependence on finite resources and improve end-of-life management. Equally important is the adoption of design principles for circular strategies that emphasize durability, reuse, repairability, and ease of recycling. These approaches ensure that plastic products are not only more sustainable in material composition but also in how they are integrated into circular value chains. This review synthesizes recent advancements in both material innovation and design thinking, highlighting how the combination of these two dimensions can extend product lifespans, enhance recyclability, minimize waste leakage, and reduce overall environmental impacts. Case studies and emerging practices illustrate how intelligent design, when coupled with innovative materials, can support closed-loop systems, and enable plastics to shift from being a linear pollutant to a regenerative resource. At the same time, systemic challenges remain, including technological limitations, market barriers, and social acceptance issues. The paper also examines enabling policy frameworks and the role of multi-stakeholder collaboration among governments, industries, researchers, and consumers in driving this transition. By integrating innovation, design, and governance, the review presents a holistic perspective on how plastics can evolve within a circular economy toward a more sustainable future.

Keywords: *Circular Economy, Material Innovations*, *Plastics*

Chemical Properties of Washed and Sieved Offshore Sand

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This study tested the chemical properties of washed and sieved off-shore sand available at Muthurajawela sales stockpile of Sri Lanka Land Development Corporation (SLLDC). Chloride, acid-soluble Sulphates and total Sulphur contents in accordance with SLS 1397 standards were tested. Testing indicated that all the key parameters were significantly lower than the allowable limit, which points to the fact that the constructions are not restrained. Decontamination by washing and sieving is effective in eliminating or removing potentially harmful salts and other impurities commonly found in marine sands, thus, limiting the chance of reinforcement corrosion, sulphate attack and structural degradation. On the whole, the experimented offshore sand shows that it can be used and applied in reinforced concrete and plain concrete use, guaranteeing durability and the structural safety of building constructions.

Keywords: Chemical Properties, Offshore Sand

Conversion of Softlogic Furniture Factory to a Net-Zero Energy Facility

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The manufacturing sector is a major contributor to global energy consumption and greenhouse gas emissions. In Sri Lanka, the heavy reliance on imported fossil fuels has led to increasing economic vulnerability and environmental stress. This study explores the feasibility of converting the Softlogic Furniture Factory into a net-zero energy facility through renewable energy integration, energy efficiency measures, and circular economy principles. The methodology combined energy audits, life-cycle cost analysis, return on investment (ROI) calculations, and employee surveys. Findings show that solar photovoltaic (PV) systems provide the highest ROI (292%) with a payback period of less than five years, while biomass boilers and LED conversions also yield strong economic and environmental benefits. Staff surveys revealed high awareness and readiness for sustainability transitions, though production staff expressed some skepticism compared to management. The study concludes that a phased transition towards net-zero is feasible, replicable, and vital for Sri Lanka's industrial sector.

Keywords: *Furniture, Net-Zero Energy,*

AI and Smart Technologies for Monitoring and Restoring Ecosystem Health

Detecting Underwater Macroplastic Pollution: A Review on AI-Based Computer Vision Methods and their Relevance to Sri Lanka's Marine Ecosystem

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The escalating crisis of marine plastic pollution, particularly macroplastic debris, poses severe threats to marine biodiversity, coastal ecosystems, and human livelihoods. Sri Lanka, ranked among the top global contributors to marine plastic waste, suffers from a lack of systematic and scalable monitoring mechanism, especially for underwater macroplastic detection. Traditional survey methods are often manual, labour-intensive, and limited in coverage. Recent advancements in artificial intelligence, specifically computer vision, present promising alternatives for automating detection and analysis of marine debris. This review paper critically examines the current landscape of AI-based approaches for underwater macroplastic detection, highlighting their methodologies, datasets, performance metrics, and limitations. While global efforts have largely emphasized surface litter and microplastic detection, underwater macroplastic monitoring remains underexplored. Moreover, Sri Lanka lacks context-specific studies and affordable, software-based AI tools that can leverage existing underwater imagery to detect macroplastic waste. By synthesizing global literature and identifying region-specific research gaps, this paper advocates for the development of lightweight, AI-powered systems tailored to the Sri Lankan coastline. Such innovations could enable continuous monitoring, inform policy decisions, and strengthen marine conservation efforts in data-scarce and resource-constrained settings.

Keywords: *Computer Vision Technique, Deep Learning, Entanglement, Plastic Ingestion, Preprocessing, Underwater Macroplastic*

AI and IoT-Driven Framework for Monitoring and Restoring of Mangrove Ecosystem Health in Coastal Sri Lanka

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Mangrove ecosystems in Sri Lanka are critical for coastal protection, biodiversity, and carbon sequestration, but face increasing threats from anthropogenic activities and climate change. Traditional monitoring methods often lack the precision and speed required to detect early-stage degradation. This study proposes an AI- and IoT-driven framework for monitoring and restoring mangrove ecosystem health in Sri Lanka, based on the global applications of AI, IoT, and Unmanned Aerial Vehicles (UAVs) in mangrove conservation. Findings show that AI techniques such as deep learning, IoT sensors, and UAVs have shown global success in species classification, degradation detection, and high-resolution monitoring. However, Sri Lanka still lacks an integrated AI- and IoT-based system for its mangrove ecosystems. Key gaps include limited localized AI models, poor technological integration, and a lack of real-time monitoring frameworks. To address these gaps, this study proposes a Sri Lanka-specific smart monitoring framework that integrates UAVs for canopy imaging, IoT sensors for root-level data, and AI analytics for real-time anomaly detection and informed decision-making. The four-layered architecture emphasizes data acquisition, secure transmission, machine learning, and an interactive dashboard for real-time monitoring. A phased implementation strategy and recommendations for community engagement, cross-sector partnerships, and policy integration are also provided. With careful piloting and stakeholder collaboration, this framework can transform mangrove conservation efforts in Sri Lanka from reactive to predictive, data-driven ecosystem management.

Keywords: *Artificial Intelligence (AI), Coastal Sri Lanka, Ecosystem restoration, Environmental monitoring, Mangroves, Internet of Things (IoT), Unmanned Aerial Vehicles (UAVs)*

Early Detection and Diagnosis of Coconut Leaf Diseases in Sri Lanka's Coconut Triangle: A Systematic Review

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This study presents a Systematic Literature Review (SLR) on the early detection and diagnosis of coconut leaf diseases in Sri Lanka's Coconut Triangle, a region critical to the nation's agricultural economy. The review systematically evaluates published research on deep learning-based models, image classification techniques, and hybrid approaches that integrate environmental data for more reliable diagnosis. To ensure methodological rigor, the study followed the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework. Relevant studies were retrieved from scholarly databases including IEEE Xplore, ScienceDirect, Springer, Scopus, and Google Scholar using predefined search strings. Strict inclusion and exclusion criteria were applied to filter peer-reviewed articles published between 2016 and 2025, ensuring relevance and reproducibility. A total of 20 papers met the criteria for final review, covering a diverse range of approaches such as Convolutional Neural Networks (CNNs), transfer learning models (VGG16, ResNet, EfficientNet), ensemble methods (CNN–SVM), and IoT-assisted systems. The synthesis of findings highlights several insights: transfer learning-based CNNs achieved over 90% accuracy, even for subtle early-stage symptoms. Hybrid approaches combining CNNs with classifiers like SVMs improved precision, reducing false positives and enhancing robustness across coconut varieties. Integrating environmental data (humidity, rainfall, temperature) increased accuracy by 6–10%, stressing the need for region-specific models in Sri Lanka's diverse agro-climatic zones. However, most systems lack large-scale field validation, mobile deployment, and consideration of smallholder farmer resource constraints. Key research gaps include limited Sri Lanka-specific datasets, inadequate focus on multi-disease co-infections, and the absence of farmer-oriented mobile and IoT tools for real-time disease monitoring. Finally, the study highlights future directions such as building larger and varietal-specific datasets, experimenting with advanced architecture (EfficientNet, Vision Transformers), and developing lightweight, offline-capable mobile applications to ensure accessibility for rural farmers. The integration of IoT sensors and climate forecasting models is also recommended to move from disease detection towards predictive and preventive farming strategies. This review contributes to the body of knowledge by not only consolidating existing research but also providing actionable insights for researchers, policymakers, and agricultural technology developers to ensure sustainable disease management and resilience in Sri Lanka's coconut sector.

Keywords: *Coconut Triangle, CNN, Deep Learning, Image classification, Leaf disease detection, Machine Learning, Support Vector Machine*

Plastic Waste Leakage in Urban River Basins of Sri Lanka: A Geospatial and Machine Learning Perspective

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Plastic waste leakage into the urban river basins has emerged as a critical environmental challenge in Sri Lanka, threatening aquatic ecosystems, public health, and urban resilience. Rapid urbanization, inadequate waste management, and poor drainage infrastructure have accelerated the accumulation and transport of plastic debris into waterways. This study presents a geospatial and machine learning approach to identify and model the key drivers of plastic waste leakage in selected urban river basins. High-resolution geospatial datasets, including land use, population density, slope, rainfall patterns, and proximity to waste disposal sites, were integrated with field-based leakage observations. A Random Forest classifier was employed to predict leakage hotspots, achieving a moderate level of accuracy. Feature importance analysis highlighted waste site proximity, urban density, and hydrological factors as the dominant predictors. The confusion matrix further illustrates the model's strengths in identifying high-risk zones; however, misclassification in low-risk areas suggests potential improvement through the inclusion of additional environmental and socio-economic variables. The findings offer actionable insights for urban planners, waste management authorities, and policymakers to develop targeted interventions for mitigating plastic waste in riverine environments.

Keywords: *Environmental modeling, Geospatial analysis, Machine learning, Plastic waste leakage, Urban River basins in Sri Lanka*

A Systematic Review of Clinical Strategies for Bloodless Surgery: A Machine Learning Perspective

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Bloodless surgery has emerged as a critical advancement in contemporary medicine, offering safe and effective alternatives for patients who refuse blood transfusions due to religious convictions, cultural norms, personal beliefs, or in situations where resources are limited. These approaches are not merely substitutes but, when carefully implemented, have demonstrated clinical outcomes comparable to, and in some cases exceeding, those achieved through traditional transfusion-based methods. The existing literature highlights a range of transfusion-avoidance strategies, including intraoperative cell salvage, acute normovolemic hemodilution, erythropoietin therapy, and minimally invasive surgical techniques. However, current research remains largely fragmented, with many studies focusing on specific interventions or case reports, often relying on qualitative insights rather than comprehensive, data-driven analyses. The fusion of knowledge in this field is made more difficult by the variety of surgical specialities, which include trauma, gynaecological, orthopaedic, and cardiovascular care. Differences in patient outcomes, institutional protocol variations, and ethical considerations add complexity, highlighting the necessity of a cohesive, framework grounded in evidence to direct clinical judgement. To fill this gap, the current study identifies, categorises, and codes transfusion-free surgical interventions through a systematic review of peer-reviewed and evidence-based sources. 22 unique methods were found and categorised after 68 pertinent documents were reviewed. The study used sophisticated machine learning (ML) and natural language processing (NLP) techniques to extract, organise, and code treatment processes in a methodical manner for scalability, reproducibility, and accuracy. In addition to standardising current clinical knowledge, this computational method makes it possible to incorporate it into educational materials and decision-support systems. The results support both practical surgical planning and ethical considerations by offering doctors, educators, and policymakers a comprehensive resource for implementing bloodless surgery. In the end, this research establishes the foundation for data-driven, scalable models that improve patient-centered care, encourage creativity, and advance the widespread use of transfusion-free surgical techniques.

Keywords: *Bloodless Surgery, Clinical Decision Support, Natural Language Processing (NLP), Machine Learning (ML), Transfusion Avoidance*

Towards a Socio-Techno-Ecological Framework (RESCUE) for Smart Waste Management for Urban Areas in Sri Lanka: A Conceptual Model

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Urbanization in Sri Lanka is accelerating rapidly, with nearly 47% of the population projected to live in urban centers by 2035, up from 19% in 2000. This surge, coupled with rising consumerism, has intensified municipal solid waste (MSW) challenges, particularly in cities like Colombo, which generates over 1,200 tons of waste daily. Current waste management practices remain linear, collection, disposal, and landfill, leading to weak recycling rates, poor infrastructure, and limited citizen engagement, thereby worsening plastic pollution, biodiversity loss, and climate-linked disasters. This study proposes RESCUE (Real-time Environmental Surveillance and Community-Utilized Ecology), a novel socio-techno-ecological framework tailored for Sri Lankan urban contexts. The study employed a conceptual systems design approach integrating a comprehensive literature review, stakeholder mapping, cultural profiling, ecological assessment, and iterative framework modelling. RESCUE integrates AI-driven predictive analytics, IoT-enabled infrastructure, community engagement, and ecological risk profiling to deliver an adaptive, inclusive, and resilient urban waste management system. The framework comprises four interconnected layers: data infrastructure, analytics and decision-making, community engagement, and governance feedback. Comparative analysis with global systems, such as those in Barcelona, Seoul, Singapore, and INSEE's Ecocycle, demonstrates RESCUE's novelty in embedding social inclusivity, ecological sensitivity, and technological integration. Pilot deployments, multi-stakeholder collaborations, and real-time monitoring strategies are recommended for validation. This framework aligns with UN Sustainable Development Goals (SDGs), particularly SDG 11 (Sustainable Cities) and SDG 12 (Responsible Consumption). RESCUE offers a scalable, replicable model for urban waste governance in Sri Lanka and other rapidly urbanizing South Asian cities.

Keywords: *AI-driven analytics, RESCUE framework, Smart cities, Sustainability, Urban waste governance*

UTAUT-Based Predictive Analysis of Youth Adoption of Gamified Environmental Sustainability Apps

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This study investigates factors influencing the adoption of a gamified mobile application for environmental sustainability activities among Open and Distance Learning Software Engineering undergraduates in Colombo District, Sri Lanka, using the Unified Theory of Acceptance and Use of Technology (UTAUT) framework. Promoting environmental awareness requires active engagement, and gamified mobile applications can enhance participation through digital incentives and rewards. A structured questionnaire based on UTAUT constructs performance expectancy, effort expectancy, social influence, and facilitating conditions was administered to students, and predictive analysis was conducted to identify key determinants of adoption intention. Findings indicate that performance expectancy, social influence and facilitating conditions significantly influence students' willingness to use the mobile application for environmental activities. The proposed model demonstrates the potential of combining gamification with UTAUT-driven adoption strategies to foster long-term environmental stewardship among youth, highlighting the effectiveness of predictive analysis in promoting sustainable engagement.

Keywords: *Environmental awareness, Gamification, Mobile application, Sustainable engagement, UTAUT*

IoT-Enabled Real-Time Water Quality Monitoring for Shrimp Farming: Trends, Challenges, and Future Directions in Sri Lanka: A Systematic Review

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Shrimp farming plays a significant role in Sri Lanka's aquaculture industry, contributing substantially to the national economy. However, many shrimp farms still rely on manual monitoring methods to assess critical water quality parameters such as pH, temperature, dissolved oxygen, and salinity. These outdated approaches often result in delayed responses to unfavorable water conditions, leading to reduced productivity, increased shrimp mortality, and environmental degradation. This research proposes the development of an intelligent, real-time water quality monitoring system specifically designed for small- and medium-scale shrimp farms in Sri Lanka. The system integrates low-cost, semiconductor-based IoT sensors to continuously monitor key parameters and transmit the data to a cloud-based platform for real-time analysis and visualization. By automating the monitoring process, the system enables farmers to respond quickly to changes in water quality, improving shrimp survival rates and overall farm efficiency. The proposed solution is designed to be user-friendly and cost-effective, making it accessible to local farmers with limited technical expertise. This innovative approach not only enhances productivity but also promotes sustainable aquaculture practices by optimizing resource usage and minimizing environmental risks.

Keywords: *IoT, Predictive Analytics, Real-Time Water Quality Monitoring, Shrimp Farming, Smart Aquaculture*

Design and Implementation of a Low-Power Edge-WSN Architecture for Mangrove Health Monitoring in Restoration Sites

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This research presents an AI-ready, scalable, low-power wireless sensor network (WSN) framework for smart environmental monitoring in remote and ecologically sensitive coastal areas, with a focus on mangrove replantation sites impacted by plastic pollution and climate change. The system integrates custom-designed ESP32 microcontroller-based master and slave sensor nodes powered by solar energy, capable of measuring key environmental parameters including sea water level, total dissolved solids (TDS), pH, and temperature, while supporting AI-driven analytics for predictive ecosystem health assessment. Communication between nodes is achieved using LoRa on the free ISM band, employing time-slot scheduling, multi-path routing, and acknowledgment-based protocols to ensure reliable long-range, low-power data delivery. Edge computing capabilities allow on-node data preprocessing with statistical filtering, event-based reporting, and adaptive transmission to optimize bandwidth and energy consumption. A cloud-based Firebase dashboard enables real-time visualization, historical trend analysis, and integration with AI models for anomaly detection and early-warning alerts. By linking water quality monitoring with plastic pollution impact assessment and providing a robust, low-maintenance, and intelligent monitoring infrastructure, this system offers a powerful tool for advancing ecosystem restoration, sustainability management, and informed, data-driven decision-making.

Keywords: *Low-power, Smart environmental monitoring, Wireless Sensor Network*

Climate Action, Computational Modelling and Environmental Sustainability

Restoring Sri Lanka's Central Highland Cloud Forests in Bopaththalawa: A Nature-Based Fiscal Strategy for Climate Adaptation and Water Security

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Sri Lanka's Central Highlands contain one of the island's most fragile and biodiversity-rich ecosystems: the montane cloud forests. These forests provide essential services including water regulation and climate moderation, yet face increasing degradation from invasive species, frequent fires, and unsustainable land use. This study presents a community-driven restoration model implemented in Bopaththalawa by Rockland Corporate Responsibility (Olu Tropical Water), Earthlanka Youth Network, and affiliated partners to restore 23 acres of degraded cloud forest using assisted natural regeneration, native pioneer species planting, invasive species removal, and establishment of decentralized community-run nurseries. Over a 3-year period, 128 plant and 123 animal species were recorded, with more than 34% categorized as threatened under IUCN criteria. Restoration belts showed sapling survival rates exceeding 70%, increased faunal presence, and reduced signs of degradation. Women and local community participation were integrated through home garden plant propagation, biodiversity monitoring, and forest fire prevention. The results demonstrate that low-cost, participatory models can initiate ecological recovery in cloud forest landscapes while delivering co-benefits for livelihoods and climate resilience. The approach aligns with Sri Lanka's National Adaptation Plan and Nature-based Solution (NbS) frameworks globally. Key policy recommendations include institutionalizing Payments for Ecosystem Services (PES) schemes, leveraging Corporate Social Responsibility (CSR) based restoration, and scaling decentralized restoration across montane regions of the Global South.

Keywords: *Cloud forest restoration, Community-based conservation, Nature-based solutions*

Feasibility Study on Forecasting Water Level Fluctuation of the Senanayaka Samudra Reservoir

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The Senanayake Samudraya Reservoir in Ampara District of Sri Lanka is a crucial piece of water management infrastructure. It supports various sectors, including agriculture, industry, and domestic water needs. The country experiences variable rainfall patterns throughout the year due to its tropical climate and diverse topography. Catchment area of the reservoir is 994.22 sq.km and command area is 120,254 ac (488.66 sq.km). Data for the reservoir was collected from irrigation department, Ampara and 26-year (1990-2015) rainfall and temperature data collected from Department of Meteorology and Irrigation department. This study developed a comprehensive water level forecasting system for the Reservoir to improve water resources management system. Both Hydrologic Engineering Centre-Hydrologic Modelling System (HEC-HMS) and Soil Conservation Service-Curve number (SCS-CN) methods were used to simulate the surface inflow and compared, with SCS-CN proving more reliable for this study. Then, the base flow of the reservoir was determined. The Long Short-Term Memory (LSTM) recurrent neural network (RNN) was employed to forecast water level fluctuations. The model performance was evaluated based on the computed statistical parameters. The performance of model is very good with Coefficient of Determination (R2) = 0.998, Mean Square Error (MSE)= 0.0357 and Root Mean Squared Error (RMSE)= 0.189. Finally, it can be concluded that the model can be used effectively in forecasting water level fluctuations, providing valuable insights for reservoir management.

Keywords: *HEC-HMS model, Long Short-Term Memory (LSTM), Recurrent Neural Network (RNN), Senenayake Samudra Reservoir, SCS-CN method*

Extending the European Union Deforestation-free Regulation to Sri Lanka's Land-Based Shrimp Aquaculture: Environmental, Regulatory and Trade Justifications

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Abstract The European Union Deforestation-free Regulation (EUDR), adopted in 2023, aims to ensure that commodities entering the EU are free from deforestation after 31st December 2020, with strict requirements for traceability and due diligence. Although aquaculture is not currently included, this study evaluates the environmental and regulatory dimensions of Sri Lanka's land-based shrimp farming to assess its suitability for future inclusion. Evidence from coastal districts shows that shrimp farming contributes to mangrove loss, carbon emissions, biodiversity decline, and reduced coastal ecosystem services; impacts comparable to those of EUDR-regulated commodities. Regulatory analysis reveals significant gaps in geolocation-based traceability, legality verification and land-use monitoring. While Sri Lanka's marine resource governance framework provides partial protection, it remains inadequate in preventing the conversion of critical habitats for aquaculture, and centralized Monitoring, Reporting, and Verification (MRV) systems are absent. Moreover, voluntary certifications such as ASC and BAP remain underutilized. This study recommends aligning national governance with EUDR principles through legal reforms, MRV implementation, and wider adoption of certification schemes. It also highlights ecological risks from converting regenerating mangroves on abandoned salterns, advocating for their consideration within deforestation-free compliance frameworks.

Keywords: *European Union Deforestation-free Regulation (EUDR), Land-based shrimp aquaculture, Sri Lanka, Mangrove degradation*

Integrating Biodiversity Assessments as a Complementary Framework for EUDR Implementation in High-Risk Commodity Landscapes

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The European Union Deforestation Regulation (EUDR) is a framework to minimize the environmental footprint of imported commodities to the EU by enforcing deforestation-free supply chains. While the regulation marks a significant benefit in trade-linked environmental governance, its current implementation mainly focuses on canopy loss and land-use change, with limited attention to biodiversity or habitat integrity. This narrows down the ecological scope risks excluding landscapes of high conservation value that may not meet formal forest definitions but nonetheless support rich and threatened biodiversity. We hereby argue that integrating biodiversity assessments into the EUDR compliance framework as an important factor for strengthening ecological integrity, improving traceability and aligning the regulation with global biodiversity objectives. The assessment examines a conceptual model of embedding biodiversity metrics into due diligence processes with examples of sustainability components to EUDR. It also looks at the ecological and strategy updates to the current structure, particularly for commodities that are not included yet which remain unregulated despite their documented biodiversity and deforestation impacts. The paper also discusses implementation challenges, including lack of capacity in producer countries, the risk of smallholder exclusion and the need for EU-endorsed biodiversity monitoring protocols. By embedding such a complementary pillar to deforestation-free verification, the EUDR can evolve into a more holistic and credible ecological governance instrument supporting not only forest conservation but also biodiversity protection, sustainable livelihoods and nature-positive trade systems.

Keywords: *Biodiversity Assessment, Deforestation Regulation, Ecosystem Integrity, EUDR, Trade Commodities*

Analysis of Cadmium and Chromium Temporal Variations in Muthurajawela Wetland: A Heavy Metal Contamination Assessment from Kerawalapitiya Waste Park

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This research investigates the temporal variations of cadmium (Cd) and chromium (Cr) contamination in Muthurajawela wetland, Sri Lanka's largest saline wetland ecosystem, resulting from leachate discharge from the adjacent Kerawalapitiya Waste Park. A comprehensive monitoring study was conducted over multiple seasonal periods from March 2024 to October 2024, analyzing heavy metal concentrations across twelve sampling points (SP01-H3SP12) within the wetland system. The study revealed significant spatial and temporal variability in both cadmium and chromium concentrations, with cadmium levels ranging from 0.00004 mg/L to 0.78 mg/L across different sampling locations and seasonal periods, while chromium concentrations varied from 0.00003 mg/L to 8.6 mg/L. Critical contamination hotspots were identified at sampling points SP01, D1SP02, and D2SP3, which consistently exhibited elevated heavy metal concentrations throughout the monitoring period. Seasonal analysis demonstrated that cadmium contamination peaked during the March 2024 period at several locations, while chromium showed maximum concentrations during August 2024 at specific monitoring sites. The proximity to the waste disposal facility emerged as a primary factor influencing contamination levels, with downstream sampling locations showing progressive dilution effects. These findings highlight the urgent need for enhanced leachate management strategies and continuous monitoring protocols to protect this internationally significant wetland ecosystem from persistent heavy metal contamination. The study contributes valuable baseline data for environmental protection measures and informs policy development for sustainable waste management practices in sensitive coastal wetland environments.

Keywords: *Heavy metals, Leachate, Muthurajawela Wetland, Waste Park*

Designing a Wildlife Crossing Structure to Mitigate the Impacts of the Central Expressway Stage 4 - Kurunegala to Dambulla

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This research project addresses the challenges posed by the continuous expansion of transportation networks on indigenous wildlife and their habitats, with a specific focus on mitigating the impact of the Central Expressway Stage 4 (CEP S-4) on local fauna. The approach begins with a comprehensive literature review, delving into animal habitats, wildlife behavior, and best practices for integrating wildlife crossings into transportation projects. Building on the insights gained from previous research, the project conducts wildlife assessments using field observations and surveys to identify and document various species within the study area, considering factors such as distribution, population densities, and habitat requirements. An impact assessment follows, examining the potential consequences of the proposed CEP Stage 4 on indigenous wildlife populations and their habitats. Key factors under scrutiny include habitat fragmentation, disturbances from transportation infrastructure, and barriers to wildlife movement. Armed with these findings, the project then develops a design proposal for wildlife crossing structures. Collaborating with ecologists, wildlife experts, and stakeholders ensures that these structures align with wildlife movement patterns, habitat preferences, and conservation needs. The aim is not only to reconnect fragmented habitats but also to provide safe passage for wildlife across the expressway, ultimately contributing to the conservation of indigenous wildlife populations. In essence, this research project seeks to bridge the gap between urban development and biodiversity conservation, presenting a model for sustainable transportation planning that promotes a harmonious coexistence between human-made infrastructure and the natural world.

Keywords: *Expressway, Span, Terrestrial Animals, Wildlife Crossing*

Mobile Sources Emissions to Ambient Air Quality: Case Study on the Baseline Road, Greater Colombo, Sri Lanka

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Understanding air quality near roads is important for mitigating urban air pollution and safeguarding public health. Empirical evidence has shown that vehicular emissions significantly contribute to elevated air pollution levels in near-road environments, as indicated by monitoring data collected predominantly during short-term campaigns worldwide. This study evaluates the contribution of vehicular emissions to near-road ambient air quality (PM2.5, CO, and NO2) in a significant segment of Baseline Road, of the Colombo metropolitan area, Sri Lanka. using the AERMOD view dispersion model. AERMOD was validated using monitored data for CO, NO2, and PM2.5. The results show that CO has a fractional bias (FB) of 0.421, a normalized mean square error (NMSE) of 0.294, a factor of two (FAC2) of 0.652, and a correlation coefficient (R) of 0.5. For PM2.5, the FB is 1.858, the NMSE is 32.865, the FAC2 is 0.037, and the correlation coefficient (R) is 0.376. NO2 has an FB of 0.727, while NMSE of 0.954, a FAC2 of 0.467, and a low correlation coefficient (R) of -0.407. The simulated results were under-predicted because only the exhaust emissions of vehicles were considered for the study. The maximum contribution of CO (>83%), PM2.5 (10%) and NO2 (77%) was observed near road junctions. However, the maximum concentrations of pollutants were below the ambient air quality standards stipulated by the Central Environmental Authority of Sri Lanka.

Keywords: *Air Pollution, Air Quality, Dispersion Modelling, Vehicular Emissions*

Multi-Temporal GIS-Based Analysis of Surface Water Susceptibility to Pollution in the Kala Oya Basin, Sri Lanka

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This research focused on identifying pollution-prone areas within the surface water bodies of the Kala Oya basin and developing a Surface Water Susceptibility Index (SWSI) to assess their vulnerability to contamination. The SWSI maps were developed using six main parameters: streams, soil type, ground slope, settlements, land use, and the road network. These factors were analyzed using Arc GIS 10.8 software and the real time data were obtained through sample surveys in 2019. To integrate these multiple parameters, the Multi-Criteria analysis (MCA) was conducted, using the weighted linear combination technique (WLC). This approach enabled a systematic categorization and aggregation of each criteria where its importance is ranked then integrated using expert opinion and review of related studies. The applicability of the SWSI model was further assessed using Inverse Distance Weighted (IDW) interpolated real time water quality data obtained for the Kala Oya basin. The research identified that the pollution risk was moderate in the basin area and it occupied 34.92 % of the total basin area with the lower and the upper basin having a relatively low pollution risk. From the map removal techniques, it was found out that sensitivity of surface water to pollution is more influenced by slope then by soils, road, LULC, and settlement and streams in order. This research calls for specific pollution controls geared towards the sensitive features already distinguished. These findings are evidence that support the measures of pollution control and also is a pointer to how best water quality could be managed for the best results fits most effectively the local water management and land use.

Keywords: *Multi-criteria Analysis (MCA), Pollution SWSI model, Surface water, Susceptibility, weighted linear combination (WLC)*

Revising and Optimizing Iso-Yield Curves for the Diverse Agro-Climatic Zones of Sri Lanka

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The most recent nationwide Iso-Yield curves, which illustrate regional variations in crop productivity, were last updated in Sri Lanka in 1978. This lengthy gap in data is significant, especially given the considerable changes in climate, hydrology, and agricultural practices that have taken place over the past four decades. This study aims to update Sri Lanka's yield maps for its two primary monsoon-driven cultivation seasons: Maha (Northeast Monsoon) and Yala (Southwest Monsoon). To achieve this, an integrated approach is employed, combining river gauge data, records of reservoir operational inflows and outflows, and observed agricultural production statistics. Historical and contemporary hydrological datasets from key river basins were compiled and analyzed to estimate irrigation water availability patterns across districts. These hydrological indicators were statistically correlated with crop yield performance data derived from national agricultural statistics, facilitating the spatial interpolation of updated yield distributions. Advanced geospatial analysis techniques and GIS-based modeling were employed to delineate field zones characterized by homogeneous yield potential, resulting in the generation of updated Iso-Yield curves for both monsoon seasons. The findings indicate significant spatial shifts in high- and low-yield zones compared to the 1978 maps, reflecting the combined effects of climate variability, alterations in reservoir operations, and evolving irrigation practices. The updated Iso-Yield curves provide a more accurate representation of Sri Lanka's current agricultural productivity landscape, aiding in precise zoning for optimized agronomic decisions, targeted irrigation scheduling, and region-specific crop planning. This study offers a valuable decision-support tool for policymakers, irrigation managers, and farmers, while establishing a methodological framework for future yield mapping endeavors in monsoon-dependent agricultural systems.

Keywords: *Agricultural Crop Productivity, Climate, Hydrology, Iso-Yield Curves*

National-scale Rainfall Bias Correction CMIP6 Projections Using Statistical Methods with Explicit SSP Comparison and Seasonal Regimes

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General Circulation Models (GCMs) play a vital role in forecasting future climate changes; however, their rainfall outputs frequently exhibit significant biases due to coarse resolution and oversimplified processes. Consequently, it is essential to apply bias correction methods to this data before utilizing it for climate change initiatives. This study introduces a context-aware framework that integrates multiple statistical bias correction methods specifically tailored to the climatic zones and rainfall characteristics of Sri Lanka. Rainfall data from 68 monitoring stations across the island were utilized alongside outputs from the CNRM-CM6-1 GCM. The study was conducted under moderate and high climate change scenarios, Shared Socioeconomic Pathways (SSP) (SSP2-4.5 and SSP5-8.5) from the CMIP6 program. Bias correction techniques, including Linear Scaling (LS), Empirical Quantile Mapping, Delta Change, Power Transformation, and Local Intensity Scaling, were employed to evaluate the performance of each method. The framework dynamically identifies the most effective method based on metrics such as RMSE, R^2, and Taylor diagrams. The results indicate that LS significantly enhances rainfall accuracy, particularly for monthly precipitation, outperforming other methods under SSP2-4.5. However, LS has limitations, particularly regarding its inability to account for topographic variation and spatial biases, highlighting the necessity for complementary approaches. Spatial analysis revealed regional disparities, especially in northern areas, attributed to local climate dynamics. Furthermore, the corrected data demonstrate improved accuracy and effectively capture seasonal trends, including intensified extremes projected under SSP5-8.5. This research underscores the importance of adaptive, regionally tailored correction strategies to enhance the reliability of rainfall data in climate impact assessments and water resource management.

Keywords: *Bias-correction, Climate changes, Linear scaling*

Study on Impact of Railway Bridge Piers on Flood Inundation at Adjacent Riverbanks of Kelani River, Sri Lanka

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The Kelani River basin in Sri Lanka frequently experiences severe flooding, especially in its lower catchment, due to intense rainfall over the steep upper catchment. Man-made structures such as bridges with insufficient clear spans and those located in floodplains can worsen flooding by impeding natural river flow. In particular, railway bridge piers disrupt flow dynamics and hydrological processes, potentially leading to adverse environmental and hydraulic consequences. This study develops a hydrodynamic model to assess the impact of railway bridge piers on flood inundation along the Kelani River's adjacent banks. Geometrical and hydro-meteorological data were collected, with the 2016 storm event used as a reference. Peak flow for this event was determined using the HEC-HMS model and then applied in a detailed HEC-RAS hydrodynamic simulation. Scenario-based simulations demonstrated that the Old and New Railway Bridges increased upstream flood height by approximately 0.010 meters, indicating a hydraulic obstruction. Moreover, the river's flow area was altered, showing an average increase of about 2.25 square meters, suggesting channel constriction due to the piers. These findings highlight the hydraulic impacts of railway bridge infrastructure on the Kelani River system. The slight increase in upstream flood height implies increased flood risk and reduced hydraulic capacity, while the change in flow area indicates morphological alterations. The results emphasize the importance of considering bridge-induced flow disruptions in flood risk assessment, river management, and future infrastructure design.

Keywords: *HECRAS modelling, HECHMS modelling, Kelani River floods, Kelani River Flood Inundation*

Investigating the Temporal Variations in Leachate Contamination Risk Assessment at Kerawalapitiya Waste Park within Muthurajawela Wetland, Sri Lanka: An LPI-Based Investigation

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This research examines temporal fluctuations in leachate contamination potential at Kerawalapitiya Waste Park, Sri Lanka's largest open waste disposal facility situated within the ecologically significant Muthurajawela wetland ecosystem. Using the Leachate Pollution Index methodology across three distinct seasonal periods over a comprehensive three-year monitoring campaign from 2022 to 2024, the study assessed multiple physicochemical parameters encompassing organic pollutants, inorganic components, heavy metals, major ions, and nutrient indicators to determine environmental contamination threats to the adjacent wetland system. Results revealed substantial seasonal and annual variability in contamination potential, with total LPI values fluctuating around 10.0 during the 2023 wet season and 37.2 during the 2024 dry season. Organic matter emerged as the primary pollution contributor, peaking at 80.0 during the 2024 intermediate season, while heavy metal contamination exhibited erratic behavior with maximum levels of 43.8 recorded during the 2024 dry season. The year 2024 represented the most environmentally hazardous period, with all measured components demonstrating elevated contamination levels. Seasonal analysis throughout the monitoring period revealed irregular patterns, suggesting that conventional seasonal management approaches are inadequate for addressing the complex interactions between climatic factors, waste composition, and leachate generation processes. These findings underscore the urgent need for enhanced monitoring protocols and adaptive pollution management strategies that can respond to temporal variations in contamination composition, while contributing valuable insights into leachate contamination dynamics in tropical wetland environments and informing targeted environmental protection measures for sensitive coastal ecosystems.

Keywords: *Kerawalapitiya, Leachate, Muthurajawela wetland, Pollution, Waste Park*

Modeling Microplastic Impacts on Soil Quality in Sri Lanka Using Data Analytics and Graph Data Science

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Plastic and microplastic pollution has emerged as a critical environmental challenge, with profound implications for soil ecosystems. In Sri Lanka, rapid urbanization and inadequate waste management exacerbate soil contamination, yet the specific impacts on soil quality remain underexplored. This study leverages graph data science and computational modeling to assess how plastic and microplastic pollution affects soil health, focusing on soil fertility, microbial activity, and contaminant dispersion patterns. Using a PRISMA-based systematic literature review, we identify a critical research gap: the lack of integrated, data-driven approaches to quantify microplastic soil interactions in tropical agricultural contexts like Sri Lanka. Our methodology employs graph-based networks to model pollutant interactions and computational simulations to predict long-term soil degradation. The analysis incorporates six graphical representations and five data tables to illustrate pollutant distribution, soil property changes, and microbial disruptions. Findings suggest that microplastics significantly alter soil structure and nutrient cycling, with implications for agricultural productivity. This research provides actionable insights for policymakers and environmental practitioners in Sri Lanka to mitigate soil pollution through targeted interventions.

Keywords: *Computational modeling, Graph data science (GDS), Plastic pollution, Particularly microplastics, Soil quality*

Microplastics and Water Reserves in Sri Lanka: A Spatial Hotspot and Risk Assessment Framework

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Microplastic plastic particles smaller than 5 millimeters are increasingly detected in freshwater systems, with potential effects on ecosystems, water treatment operations, and human health. In Sri Lanka, water reserves such as major reservoirs like Victoria, Randenigala, and Kotmale, as well as natural lakes such as Koggala and Bolgoda, and river-fed tanks in the dry zone, are especially vulnerable. This is largely due to rapid urbanization, tourism, aquaculture, and poor solid-waste management practices. This paper presents a nine-part framework to assess the occurrence, sources, transport, and risks of microplastics across Sri Lanka's inland waters. The proposed workflow brings together field sampling, laboratory characterization using methods such as density separation and FTIR or Raman confirmation, and geospatial hotspot modeling through techniques like Getis–Ord Gi* and kernel density estimation. These are further integrated with hydrological drivers and human pressures. The framework concludes with a tiered risk index designed to help prioritize interventions for drinking-water intakes, irrigation tanks, and recreational lakes.

Keywords: *Hydro Spatial Statistics, Hotspot analysis, Microplastics, Water risk assessment, Water security*

GIS Hotspot Risk Scoring Model for Plastic Leakage in Urban Lakes in Sri Lanka

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Plastic pollution has emerged as a critical environmental challenge worldwide, with developing nations like Sri Lanka facing acute impacts in urban aquatic ecosystems. This study develops a Geographic Information Systems (GIS)-based hotspot risk scoring model to identify and prioritize plastic leakage zones in urban lakes of Sri Lanka, focusing on areas prone to waste accumulation due to anthropogenic activities. Utilizing Multi-Criteria Decision Analysis (MCDA) and weighted overlay techniques within a GIS framework, the model integrates diverse spatial datasets, including drainage inflows, human settlements, road networks, vegetation cover, tourism activity, and waste management infrastructure. Each factor is normalized, weighted based on expert insights and empirical evidence from pollution pathways, and overlaid to produce a composite risk index. The resulting risk map classifies areas into five categories: Very Low, Low, Medium, High, and Very High. Validation through field observations and sensitivity analysis confirms the model's robustness, with a strong correlation ($R^2 = 0.89$) between predicted risks and observed litter counts. The outputs provide actionable insights for targeted interventions, such as enhanced waste collection and public awareness campaigns, to mitigate pollution in urban waterways. This approach not only addresses local plastic leakage but also contributes to broader sustainable development goals in managing marine and freshwater pollution.

Keywords: *Environmental management, Plastic pollution, Multi-Criteria decision analysis, Risk mapping, Weighted overlay*

Using Graph Data Science to Assess the Impact of Plastic Pollution on Air Quality in Sri Lanka

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Plastic pollution and air pollution are two pressing global challenges that have often been studied separately, leaving their interactions underexplored. Mismanaged plastic waste not only contaminates land and water but also degrades air quality through open burning and airborne microplastics. Sri Lanka generates over 1.5 million tons of plastic waste annually yet recycles only ~3%. This low recycling rate contributes to severe pollution from unmanaged plastics. Meanwhile, urban air quality has deteriorated, with smog episodes in Colombo often exceeding an Air Quality Index (AQI) of 150. This study assesses the impact of plastic pollution on air quality in Sri Lanka using a combination of systematic literature review and graph data science techniques. This study applied a PRISMA-based systematic review of global and local studies to evaluate the contribution of plastic waste to air pollution. In parallel, graph data science modeling was applied to connect key entities (plastic waste, burning practices, microplastics, air quality indicators, health outcomes) and to identify critical linkages. The literature review revealed that open burning of plastic waste releases toxic particulate matter and gases (e.g. dioxins, furans), and recent studies have detected microplastics in ambient air, posing inhalation risks. Our graph analysis of Sri Lankan data suggests a strong correlation between increasing plastic waste and rising ambient $PM_{2.5}$ levels, and highlights "open burning" of waste as a central node linking plastic pollution to poor air quality. The integrated graph model identified that reducing open waste burning and plastic leakage could significantly improve air quality. We conclude that plastic pollution is not only a marine or soil issue but a major air quality threat, particularly in developing countries. Mitigating plastic pollution (through better waste management, reduced single-use plastics, and prevention of burning) can yield co-benefits for air quality and public health. The novel application of graph data science in this context demonstrates a powerful approach to uncovering complex relationships in environmental systems, filling a critical research gap and informing more holistic pollution control strategies.

Keywords: *Airborne microplastics, Ambient air pollution, Graph data science, Open waste burning, Plastic waste management*

Spatial and Temporal Patterns of Riverine Plastic Waste Inputs to the Ocean in Sri Lanka: Integrating Waste Management Archetypes and Hydrological Drivers

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Understanding the plastic waste input to the ocean is a critical study to keep the Indian ocean clean, The research problem addressed in this study is to quantify, analyze, and visualize the spatial and temporal patterns of plastic waste inputs from river mouths into the ocean in Sri Lanka. This involves integrating geospatial and tabular data on riverine plastic pollution, identifying high-contributing river mouths, examining seasonal trends, and relating these patterns to waste management archetypes. The ultimate goal is to inform targeted interventions for reducing plastic leakage from land-based sources into marine environments. This research would analyze the spatial distribution and seasonal trends of plastic waste entering the ocean from Sri Lankan rivers, using geospatial and statistical methods. It would also examine how different waste management archetypes contribute to plastic leakage, and assess the influence of hydrological variables (e.g., runoff) on plastic input variability. The findings could inform targeted interventions for reducing plastic pollution at critical river mouths and improving waste management strategies. The research integrates Geographic Information Systems (GIS), remote sensing data, hydrological analysis, and field sampling to identify high-risk zones. We have employed Getis-Ord Gi* and Kernel Density Estimation (KDE) to identify statistically significant clusters. The primary results show strong correlations between plastic accumulation and anthropogenic activities including urban runoff, tourism, and inadequate waste management. The findings provide an evidence-based framework for targeted interventions and sustainable policy development.

Keywords: *Computational modeling, Plastic pollution, Remote sensing, Spatial hotspot modelling, Waste management*

Groundwater Modelling to Investigate the Oil Spill Contamination of the Chunnakam Aquifer in Jaffna Peninsula

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In the Jaffna peninsula groundwater is the only water source for sustaining life and environment. Chunnakam aquifer has the largest capacity and acceptable quality for drinking and other usages among main four aquifers in Jaffna peninsula. Over extraction of groundwater and anthropogenic activities affected the quality of ground water in different ways. In 2013, continuously fuel smell has been observed in Chunnakam water intake site which is situated very close to the Chunnakam fossil fuel power plant. And on analysis, existence of contamination was proved in 500m surrounding area. In order to find out the present situation of oil and grease contamination, an attempt was made through numerical modeling to analyze the contamination. The result reveals that the contamination and contaminated area is decreasing with time and along depth of aquifer. And the high oil and grease concentration layers were observed in the surrounding of Chunnakam power station. Around 375m surrounding of Chunnakam power station area having the high risk of contamination. Based on our site investigation we could see the oil and grease layers with naked eyes in several wells. Collected oil and grease data contains high concentration of 7.6mg/l.

Key words: *Chunnakam aquifer, Groundwater modeling, Oil and Grease, Power station.*

Public Awareness and Educational Initiatives

The Missing Link in Sustainability: Cultivating Eco-Love as a Tool for Behaviour Change in Plastic Use

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Plastic pollution has emerged as one of the most urgent environmental crises of our time. While global efforts have intensified to curb plastic use, many interventions still rely heavily on cognitive awareness, regulatory enforcement, and fear-based messaging. These approaches, although informative, often fail to achieve sustained behavioural change. This study introduces the concept of Eco-Love, a deep emotional bond with the natural world, as a sustainable motivator for reducing plastic consumption. Drawing from positive psychology, emotional intelligence, the biophilia hypothesis, and self-determination theory, the objective is to examine how fostering affectionate connectedness with nature can generate intrinsic and enduring environmental behaviours. The research adopts a Systematic Literature Review (SLR) of twenty-eight empirical studies published between 2008 and 2023, focusing on emotional and psychological drivers of pro-environmental behaviour, particularly plastic use reduction. The findings, analysed thematically, reveal three recurring insights: the limitations of fear-based campaigns, the role of nature connectedness and emotional immersion, and the predictive value of emotional intelligence for sustainable practices. Building on these insights, the study develops the Eco-Love Behaviour Change Model, which integrates positive emotions as catalysts, emotional intelligence as a mediator, biophilia as a foundation, and self-determination theory as a motivational bridge. This conceptual framework forms the basis for transformative sustainability education programs. The paper concludes with practical recommendations for embedding Eco-Love into public awareness campaigns, school curricula, corporate training, and community-based initiatives. By shifting sustainability from obligation to affection, Eco-Love reframes environmental protection as a personal, values-driven commitment rather than a compliance-based task.

Keywords: *Environmental law, Legal reform, Plastic waste, Waste regulation*

A Theory Synthesis and Systematic Literature Review on Cognitive Biases in Climate Shaped Plastic Behaviors

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This study presents a theory synthesis and systematic literature review exploring how climate-related emotions particularly eco-anxiety and solastalgia interact with cognitive biases to perpetuate unsustainable plastic consumption, even in the face of increasing environmental awareness. Addressing a significant gap in sustainability science, the research investigates the psychological mechanisms behind this paradox and evaluates evidence-based intervention strategies. Methodologically, the study integrates a PRISMA-ScR guided systematic review of 94 peer-reviewed studies published between 2010 and 2024 across psychological, environmental, and policy domains. Studies were selected based on strict inclusion criteria focusing on plastic behaviors, cognitive biases, and climate affect. In addition, a novel theory synthesis is conducted, combining dual-process theory, ecopsychology frameworks and behavioral economics to develop the model "Climate-Shaped Cognition Model (CSCM) of Plastic Behaviors". Key findings reveal that eco-anxiety intensifies present bias (aggregate OR = 2.1 in 78% of studies), while moral licensing significantly increases after climate-related actions (3.2× stronger effects). Effective interventions identified include affect calibrated nudges, which result in an 18–34% reduction in plastic use, and temporal bridging tools, with an effect size of $d = 0.57$. The synthesis highlights a neurocognitive double bind, where climate emotions simultaneously heighten environmental concern and reinforce bias driven plastic behaviors. The proposed CSCM introduces three principles for intervention design: cognitive-affective alignment, dual-process engagement, and culturally contextualized implementation. This study offers the first comprehensive framework explaining how climate emotions shape cognitive barriers to plastic sustainability and provides actionable strategies for policymakers and behavior designers. It emphasizes the urgent need for standardized measures of plastic related climate distress and rigorous cross cultural evaluation of adapted interventions.

Key words: *Cognitive biases, Climate emotions, Eco-anxiety, Environmental behavior, Plastic consumption, Sustainability*

Promoting Pro-Environmental Behavior in Sri Lanka: A Literature-Based Conceptual Framework Integrating Psychology and Cultural Values

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Plastic pollution and environmental degradation are urgent challenges in Sri Lanka, threatening health, biodiversity, and sustainability. While regulatory measures exist, community-level behavior change remains underexplored, especially when framed through Sri Lanka's cultural values. This study develops a conceptual framework that integrates global psychological theories with local cultural traditions to promote pro-environmental behavior. A narrative literature review was conducted, drawing on peer-reviewed articles, policy reports, and cultural texts published between 2015 and 2025. Sources were selected for their relevance to environmental behavior, psychology, and Sri Lankan cultural practices, ensuring a balance between global and local perspectives. The framework highlights mediators such as attitudes, social norms, perceived behavioral control, eco-anxiety, and collective efficacy, anchored in religious ethics of compassion, stewardship, and interconnectedness. Practical illustrations include plastic reduction initiatives in temples, schools, and local councils. The framework offers pathways for empirical testing through surveys, pilot interventions, and community-based projects. Recommendations include culturally sensitive campaigns, youth-led initiatives, and community partnerships to strengthen collective responsibility for sustainability.

Keywords: *Cultural values, Pro-environmental behavior, Psychology, Sri Lanka, Sustainable development*

Understanding the Continued Use of Plastic Bags among the Adults over 30 years old in Sri Lanka: Patterns, Impacts, and Alternatives

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Despite regulatory bans and widespread awareness campaigns, plastic bags remain in heavy use across Sri Lanka. This study investigated the usage patterns, environmental awareness and adoption of alternatives among adults over 30 in the districts of Kandy, Kegalle, Kalutara and Hambantota. A structured questionnaire survey was conducted among 100 respondents, complemented by expert inputs and analyzed using descriptive statistics. Results revealed that 26% of participants use plastic bags daily and 36% use them 1–2 times per week, with the highest dependence among the 40–59 age group. Further, the 60–69 age group reported lower dependency and higher eco-consciousness. While 88% of respondents acknowledged marine and wildlife risks and 83.8% recognized microplastic-related health hazards, yet this awareness did not consistently translate into reduced use. Encouragingly, 85% supported investing in reusable alternatives and over 70% favored policy measures such as discounts and bans. By addressing this gap, the study highlights the mismatch between awareness and practice among older Sri Lankan consumers, pointing that awareness alone is insufficient for behavioral change; instead, a combined strategy of accessible alternatives, incentive-driven interventions and policy enforcement are needed to reduce plastic bag dependency and encourage sustainable consumer practices in Sri Lanka.

Keywords: *Adults, Alternatives, Impacts, Patterns, Plastic bags*

Relationship between Environmental, Social and Governance (ESG) corporate performance: Sri Lankan context from 2017 to 2021

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Environmental, Social, and Governance (ESG) is a non-financial reporting approach consisting of communicating the sustainability of the environment, social, economic and governance activities of the businesses. ESG is a prominent topic with lots of research worldwide and has become a critical tool towards the success of businesses. However, ESG reporting is relatively new in the Sri Lankan context, exhibiting fewer studies. Therefore, this study seeks to measure the relationship between the ESG reporting and corporate performance of the Sri Lankan companies listed under the Colombo Stock Exchange from 2017 to 2021. The analysis has incorporated the publicly available annual and sustainability reports of the companies during the time period. The ESG is quantified using the 17 categories under the Refinitiv methodology, while return on Equity (ROE) and return on assets (ROA) are used as the measures of corporate performance. The quantitative findings indicate a lack of significant correlation between corporate performance and the quantified ESG values. External factors such as the Covid-19 pandemic and economic crisis may have likely influenced this relationship. Meanwhile the internal factors including greenwashing and inadequate active integration of sustainability towards the business operations may also have highly impacted. The study highlights the need for improved ESG frameworks and more connection of sustainability practices with corporate operations in Sri Lanka. These findings contribute to a greater discussion on ESG's role in improving company performance, remarkably in the developing countries.

Keywords: *Corporate performance, ESG rating, Return on equity, Return on assets, Refinitiv, Sri Lanka, Sustainability*

From Waste to Wardrobe: Awareness Occasion Wear Collection Crafted from Recycled Plastics, Fabric Scraps, and Leftover Materials inspired by Goddess Gala

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Plastic and textile waste from the fashion industry are major contributors to environmental pollution, affecting ecosystems and communities worldwide. Addressing this urgent issue, this paper presents "From waist to wardrobe: a sustainable awareness occasional wear fashion collection" created from recycled plastics, fabric scraps, and leftover threads that would otherwise contribute to the growing waste problem. Inspired by Goddess Gaia, the Greek mythological "Mother Earth" who embodies the living, interconnected nature of our planet, the collection narrates how human activities such as pollution, deforestation, and overconsumption endanger the Earth's health. K-Pop style dress silhouettes are used to achieve fashion trend.By merging sustainable design with cultural influence, the fashion collection shows how fashion can transcend aesthetics to become a powerful medium for raising awareness, sparking dialogue, and driving real-world action against environmental degradation.

Keywords: *Environmental awareness, Celebrity advocacy, K Pop Style, Plastic waste, Sustainable fashion*

Influence of Rewards and Gamified Experience in Waste Management Among School Students in Jaffna College Sri Lanka

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This research presents a pilot study exploring the use of rewards and gamification to enhance plastic waste management participation among school students. Conducted at Jaffna College (Vaddukoddai, Sri Lanka), the study introduced a competitive “Waste Bank” program where classes (525 students from 18 classes Grades 6–11) competed to collect polyethylene terephthalate (PET) and high-density polyethylene (HDPE) plastic bottles over a one-week period. The initiative featured gamified elements, such as a points-based competition and tangible rewards for the top-performing classes. Student engagement was enthusiastic, with thousands of plastic bottles collected, surpassing expectations. The top 10 winning classes received prizes like sports gear, snacks, and school supplies, providing immediate incentives to reinforce positive behavior. This aligns with broader research showing that gamified challenges and rewards significantly increase student involvement sustainability efforts. Students collected a total of 13,933 plastic bottles, with the top-performing class collecting 4,241 bottles (114.6 bottles per student). The study reveals how a competitive, fun atmosphere can foster environmental awareness, while also highlighting practical challenges like storage and manpower limitations. Although the study was limited to one week of data, the results suggest that gamification can be an effective tool for cultivating recycling habits in schools. The research concludes with recommendations for scaling up such programs in other educational settings, alongside strategies for overcoming logistical challenges. This research aims to demonstrate the potential for gamified approaches to instill long-term sustainable practices in young people, offering valuable insights for future waste management initiatives in schools.

Keywords: *Education, Gamification, Plastic Waste, School, Sustainable*

Assessment of Technical Sustainability as a Crucial Determinant of Long-Term Sustainability in Community-Based Water Supply Schemes in the North Central Province in Sri Lanka

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Community-Based Water Supply Schemes (CBWSS) remain central to ensuring safe and reliable water access in Sri Lanka's North Central Province (NCP), where centralized infrastructure is sparse and public health challenges, particularly Chronic Kidney Disease of unknown etiology (CKDu), persist. This study undertakes a critical, multi-dimensional sustainability assessment of 25 CBWSS in Anuradhapura and Polonnaruwa districts using 22 indicators spanning environmental, social, economic, and technical domains. A mixed-methods approach incorporating structured questionnaires, Likert-scale scoring, focus group discussions (FGDs), key informant interviews (KIIs), and field inspections was employed. Results reveal that while social participation and environmental awareness remain moderately strong, technical sustainability emerges as the most decisive factor influencing long-term system viability ($r = 0.78$, $p < 0.01$). Deficiencies in preventive maintenance, operator training, water quality surveillance, and spare-part availability significantly undermine scheme performance, leading to frequent breakdowns and, in some cases, scheme collapse or transfer to external authorities. Case study evidence further validates the central role of technical management in sustaining functionality. The study argues that strengthening operator capacity, embedding preventive maintenance culture, institutionalizing sustainability monitoring tools, and integrating technical sustainability into national policy frameworks are crucial to ensuring resilient CBWSS. Beyond local relevance, the findings contribute to global discourse on rural water governance by foregrounding technical sustainability as a cornerstone of safe and equitable water provision.

Keyword: *Community-Based Water Supply Schemes; Technical Sustainability; Rural Water Management; Chronic Kidney Disease (CKDu); Sri Lanka; Sustainability Framework*

Plastic Pollution and Protecting the Right to a Clean and Healthy Environment in Sri Lanka

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Plastic pollution has emerged as one of the most pressing environmental challenges globally, with significant ecological, health, and socio-economic implications. In Sri Lanka, the rapid increase in plastic waste, driven by urbanization, consumer behavior, and inadequate waste management systems, poses a serious threat to the environment and the well-being of present and future generations. This paper examines the issue of plastic pollution in Sri Lanka through the lens of human rights and environmental rights, focusing on the constitutional and international recognition of the right to a clean and healthy environment. It evaluates the existing legal and policy frameworks governing plastic use and waste management, identifying key gaps in enforcement, regulatory coherence, and public participation. The research employs qualitative research methods, utilizing primary data such as legislative enactments, conventions, and case laws, as well as secondary sources including journal articles and textbooks. Drawing on comparative legal approaches and sustainable development principles, the study proposes legal and institutional reforms to strengthen environmental governance and enhance the protection of environmental rights in the context of plastic pollution. This paper provides recommendations for policymakers aimed at strengthening the protection of the right to live in a clean and healthy environment, while also enhancing public awareness of this fundamental right. It argues that the recognition and effective implementation of this right are crucial for advancing environmental justice and achieving long-term sustainability in Sri Lanka.

Keywords: *Awareness, Human Rights, Plastic pollution, Right to Clean and Healthy Environment, Sri Lanka,*

Eco-Driven Community Education for Sustainable Plastic Use: Colombo Case Study

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Plastic pollution represents a severe and escalating environmental and social challenge in urban areas such as Colombo, Sri Lanka, where rapid urbanization, population density, and inadequate waste management infrastructure contribute to high volumes of mismanaged plastic waste. This results in marine ecosystem degradation, soil contamination, microplastic infiltration into food chains, and disproportionate health and economic burdens on marginalized communities. This study assesses the effectiveness of community-based educational campaigns in fostering sustainable plastic consumption behaviors among urban households, employing a sociological framework to analyze behavior change through mechanisms like social learning, community cohesion, and cultural integration. Utilizing a mixed-methods approach, the research encompassed pre- and post-intervention surveys of 150 households across two socioeconomically diverse neighborhoods Wellawatta (middle-income) and Mattakkuliya (low-income) supplemented by semi-structured interviews with 20 community leaders and residents. The intervention involved a four-week program featuring interactive workshops, localized social media campaigns in Sinhala and Tamil, and participatory clean-up drives, tailored to incorporate Sri Lankan cultural elements such as Buddhist principles of non-harm (ahimsa) and communal harmony to boost engagement and receptivity. Quantitative outcomes demonstrated a 25% overall reduction in single-use plastic consumption (from an average of 10 bags per week per household to 7.5), a 71% increase in reusable item adoption (from 35% to 60% of households), and an 83% rise in regular recycling participation (from 30% to 55%), alongside a 50% improvement in environmental awareness scores (from a mean of 2.8 to 4.2 on a 5-point Likert scale). Qualitative findings illuminated the pivotal role of peer modeling and collective efficacy in driving change, while underscoring persistent barriers including economic constraints, infrastructural deficiencies, and gender-based disparities in waste management responsibilities. Socioeconomic differences were pronounced, with middle-income areas achieving a 28% reduction compared to 22% in low-income zones, highlighting the moderating influence of resource access. The results affirm the potential of sociologically grounded interventions to advance circular economy principles such as reduce, reuse, and recycle and align with United Nations Sustainable Development Goals (SDGs) 12 (Responsible Consumption and Production) and 13 (Climate Action). By providing empirical insights into scalable, community-led strategies, this Colombo case study contributes to global and regional discourses on mitigating plastic pollution, emphasizing the integration of social sciences in environmental policy and practice for long-term behavioral and systemic transformations.

Keywords: *Behavior change, Community education, Plastic pollution, Sustainable consumption*